ECET
250E

Linear Circuits 1

Week 2 | Chapter 2: Basic Laws (With Solutions)

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Based on Fundamentals of Electric Circuits, 2026 Release — C.K. Alexander, M.N.O. Sadiku & S.J. Fredette (McGraw Hill)

Overview

- This chapter will introduce **Ohm's law**: a central concept in electric circuits.
- **Resistors** will be discussed in more detail.
- **Circuit topology** and the **voltage and current laws** will be introduced.
- Finally, **meters** for measuring voltage, current, and resistivity will be presented.

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Resistivity

- Materials tend to resist the flow of electricity through them.
- This property is called "resistance".
- The resistance of an object is a function of its length, **L**, and cross sectional area, **A**, and the material's resistivity:

$$R = \rho \frac{l}{A}$$

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Resistivity of Common Materials

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TABLE 2.1

Resistivities of common materials.

Material	Resistivity ($\Omega \cdot m$)	Usage
Silver	1.64×10^{-8}	Conductor
Copper	1.72×10^{-8}	Conductor
Aluminum	2.8×10^{-8}	Conductor
Gold	2.45×10^{-8}	Conductor
Carbon	4×10^{-5}	Semiconductor
Germanium	47×10^{-2}	Semiconductor
Silicon	6.4×10^{2}	Semiconductor
Paper	10^{10}	Insulator
Mica	5×10^{11}	Insulator
Glass	10^{12}	Insulator
Teflon	3×10^{12}	Insulator

Resistivity of Common Materials

- Resistivity also referred as Specific Electrical Resistance.
- Resistivity allows to compares the resistance of different materials at ***specified temperature*** as per their ***physical properties*** regardless of length, area of cross section.

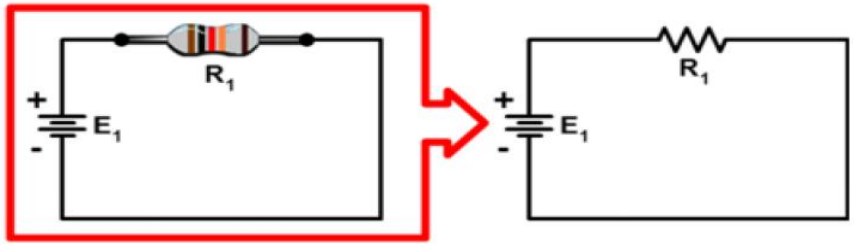
Example:

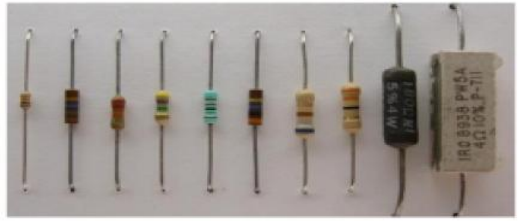
- Copper - a good conductor, has low resistance
- Glass - a good insulator, has high resistance.

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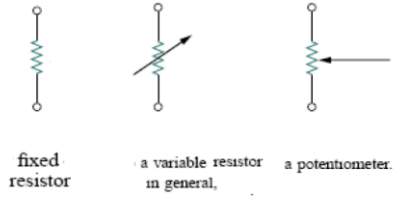
Resistor

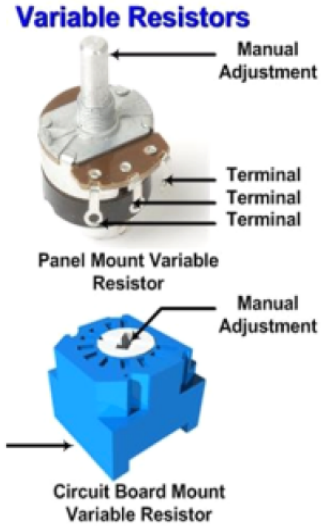
- Resistors are classified in two main categories: fixed or variable.
- Fixed resistor – whose resistance is fixed by the manufacture and does not change
  - Schematic symbol





- Variable resistor – whose resistance is adjustable from 0 to some value
  - Has three terminals
  - Two variable resistor types: potentiometer and rheostat
  - shaft for adjusting
  - Terminal 1,3 maximum, Terminal 1,2 or 2,3 adjusted resistance
  - Example: Volume control
  - Schematic symbol
  - Potentiometer – aka *Pot*
- Potentiometer Functions
  - Act as a voltage divider in the circuit





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Resistor Colour Code

- The resistor color code uses colors to represent numbers.
- Each color on the chart is equal to a number or percentage.
- Thus, these color bands provide a means to identify the resistor's resistance value and tolerance.



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③ ⓪ ①

$10 \times 10^2 \pm 5\%$

$10 \times 10^1 \pm 5\%$

$100\Omega \pm 5\%$

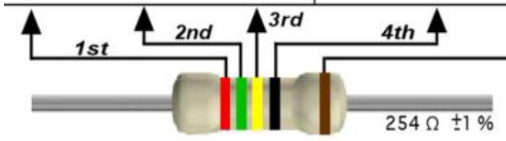
100 – 5%
100 – ?
$\frac{100 \times 5}{100} = 5$

95 → 105

4- Band Code



Color	1st Band	2nd Band	3rd Band	Decimal Multiplier		Tolerance
Black	0	0	0	1	1	
Brown	1	1	1	10	10	± 1 %
Red	2	2	2	100	100	± 2 %
Orange	3	3	3	1K	1,000	
Yellow	4	4	4	10K	10,000	
Green	5	5	5	100K	100,000	
Blue	6	6	6	1M	1,000,000	
Violet	7	7	7	10M	10,000,000	
Gray	8	8	8		100,000,000	
White	9	9	9		1,000,000,000	
Gold					0.1	± 5 %
Silver					0.01	± 10 %
None						± 20 %



5- Band Code

Calculation



First Band **Red** ---------- **2**
Second Band **Black** ---------- **0**
Multiplier Band **Yellow** -- **x10,000**
Tolerance Band **Silver** -------**10 %**

The Gold or Silver band is always placed to the right.
The resistor value is read from the left to right.

If there is no tolerance band, then find the side that has a band closest to a lead and make that the first band.

Equation

20 x 10,000 = 200,000

1,000 = 1K

Resistor = 200 K Ω
with a ± 10 % Tolerance

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Conductance

- What is conductance?
  - Measures how well a material will conduct current.
  - It is reciprocal of resistance, denoted with symbol, G
  - Measured in Siemens (S)

$$\text{Conductance} = \frac{1}{\text{Resistance}}$$

In symbol notation:

$$G = \frac{1}{R} \quad \text{(measured in siemens, S)}$$

Ω^{-1}

- Example: 1M Ohm resistance, is 10^{-6}S conductance
- An Imp. Note:
  - Large the conductance, less the resistance
  - Thus greater the conductivity

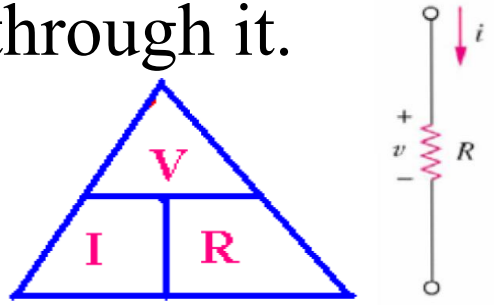
$G = \frac{1}{1M} = 10^{-6} S$

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Ohm's Law

- *In a resistor*, the **voltage** across a **resistor** is **directly proportional** to the **current** flowing through it.

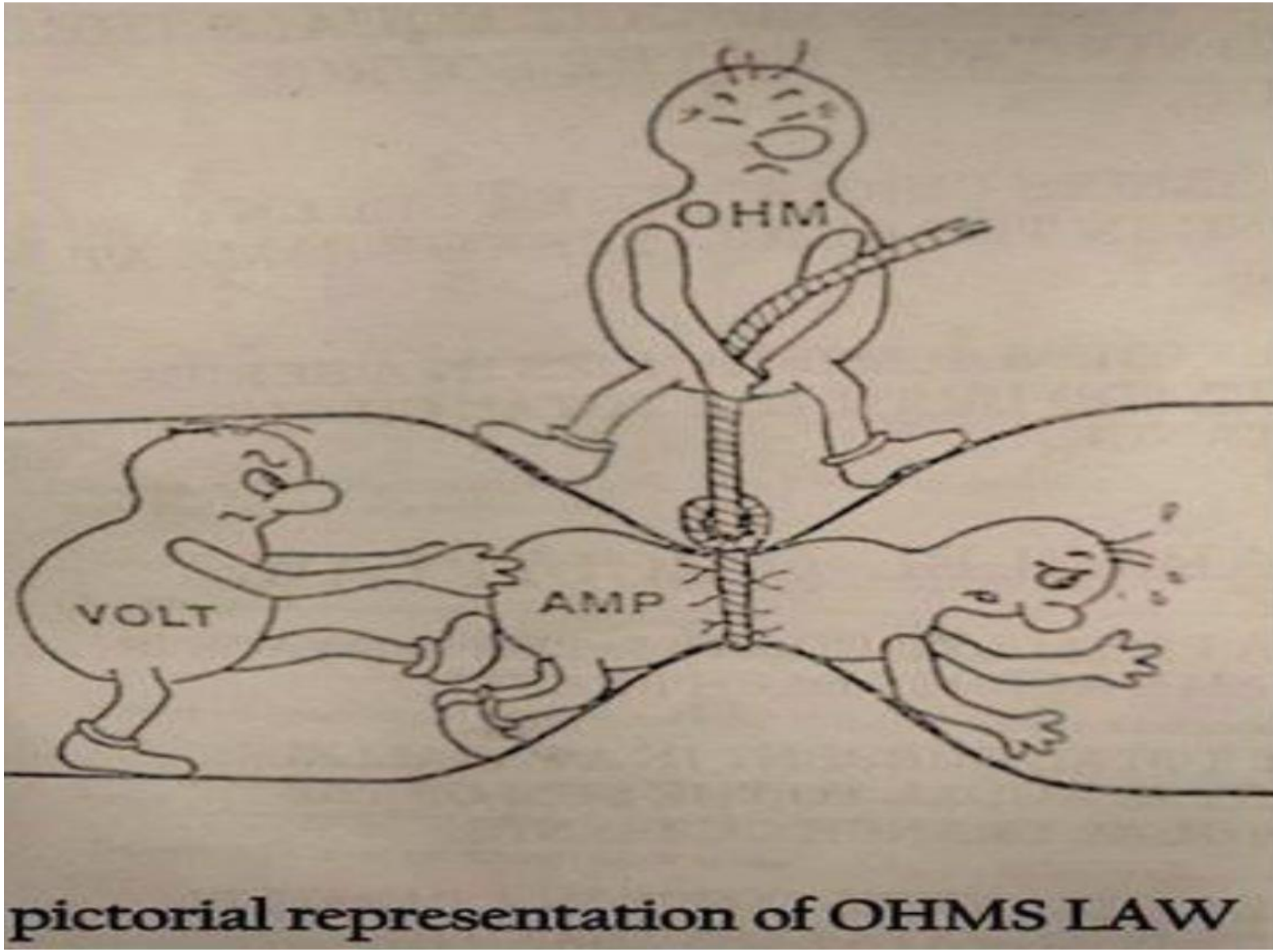
$$V = IR$$



- The resistance of an element is measured in units of Ohms, Ω, (V/A).
- The **higher** the *resistance*, the **less** *current* will flow through for a given voltage.
- Ohm's law requires conforming to the passive sign convention.

Ohm's law states that the voltage *v* across a resistor is directly proportional to the current *i* flowing through the resistor.

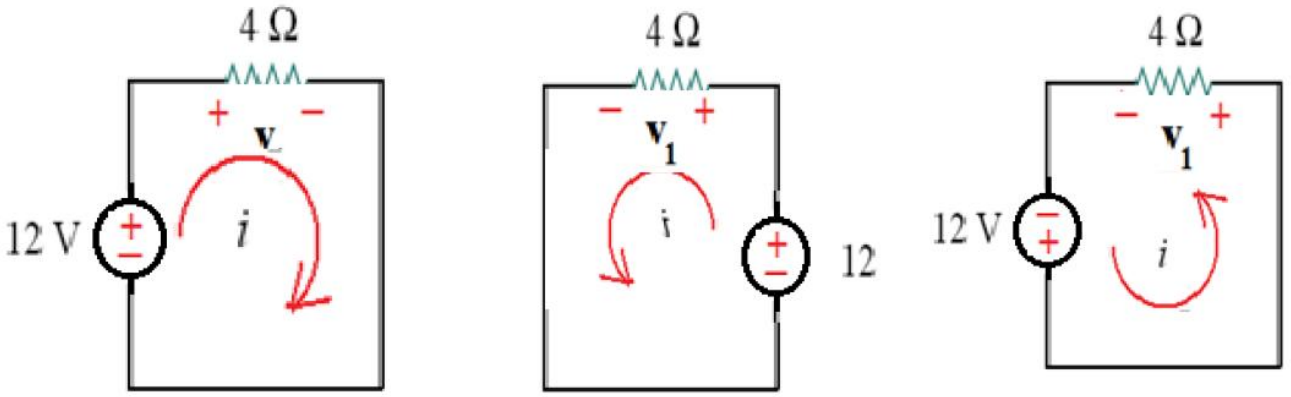
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pictorial representation of OHMS LAW

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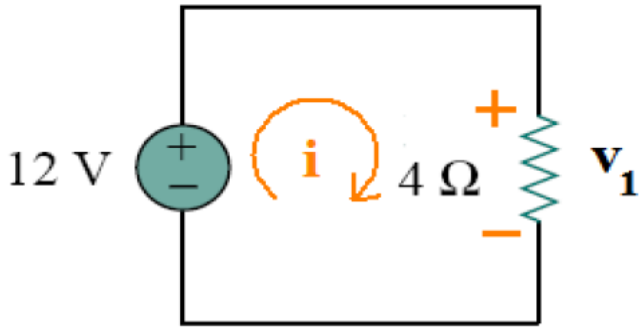
How to mark voltage and current direction on a circuit ?

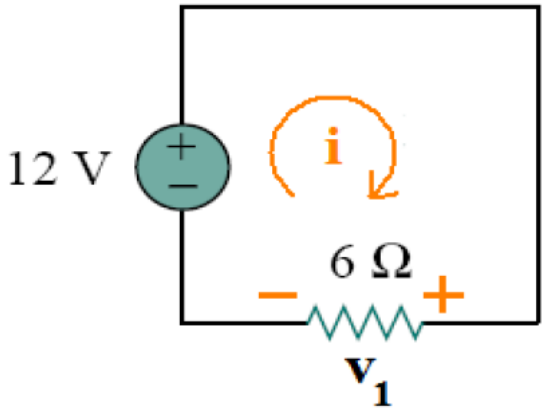


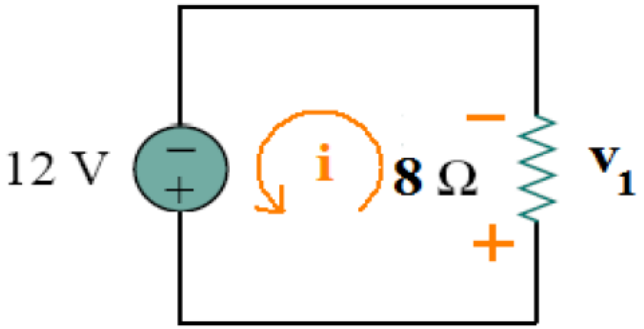
We know that the conventional current direction starts from the +ve terminal of the battery and enters the circuit.

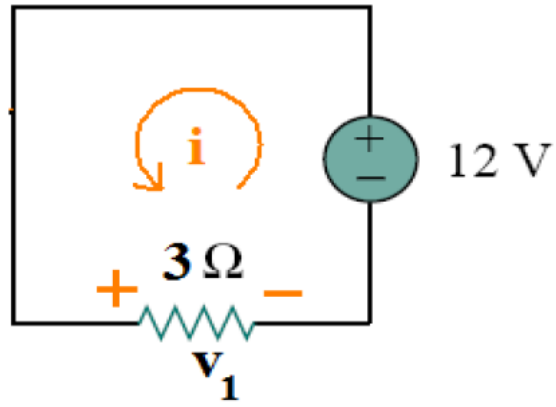
Remember: whichever end of the resistor the current touches first, we mark that end as +ve and other end -ve, for the voltage sign.

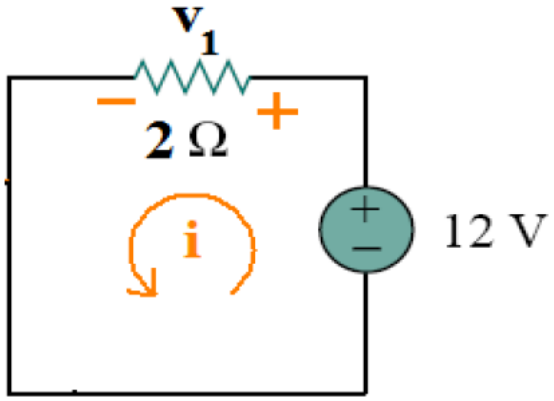
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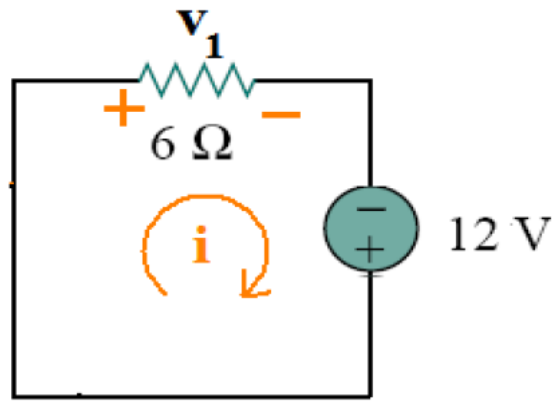










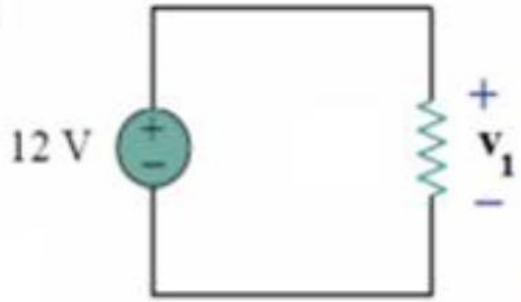


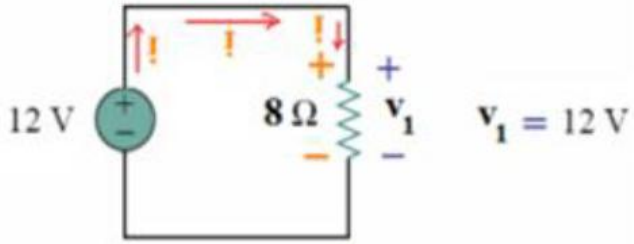
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Voltage Sign

If a voltage sign in already marked on a resistor (circuit)
we must check whether the voltage sign conforms to the current direction or not.

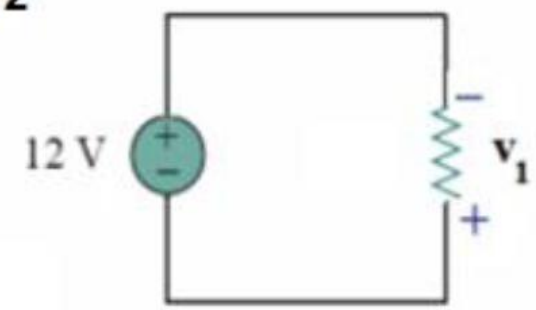
Example 1

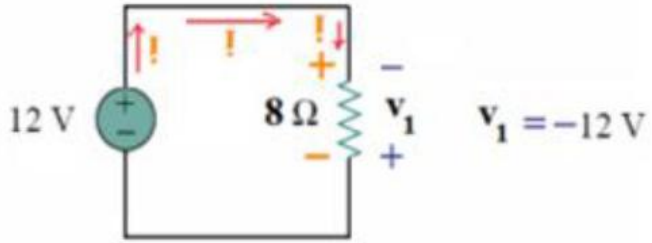




If the sign conforms to the current direction then it is OK.

Example 2





If the sign does not conform to the current direction,

the voltage value will be taken (written) as negative.

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Short and Open Circuits

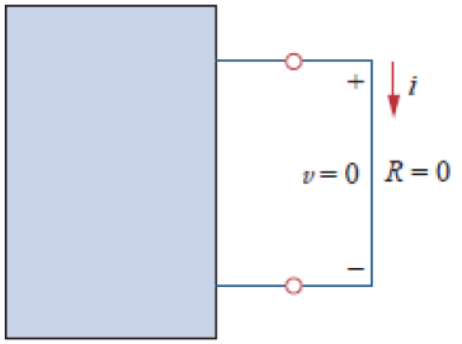
- A connection with almost **zero resistance** is called a short circuit.
- Ideally, any current may flow through the short.
- In practice this is a connecting wire.
- A connection with **infinite resistance** is called an open circuit.
- Here no matter the voltage, no current flows.

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When R = 0

A **short circuit** is a circuit element with resistance approaching zero.

$$v = iR = 0$$

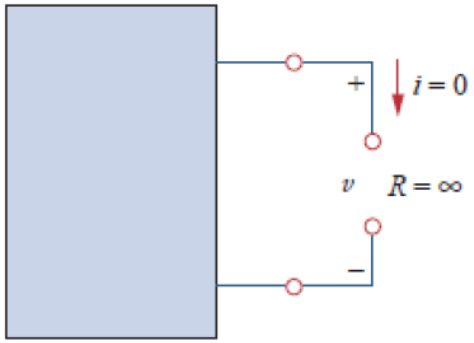


(a) Short circuit ($R = 0$)

When R = ∞

An **open circuit** is a circuit element with resistance approaching infinity.

$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$



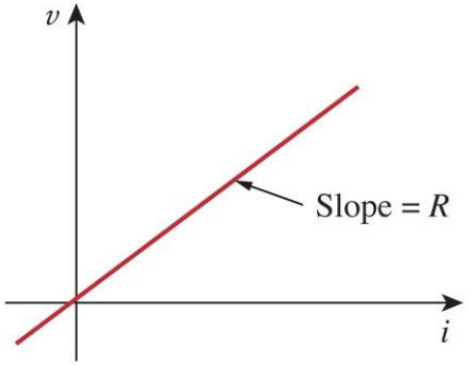
(b) Open circuit ($R = \infty$).

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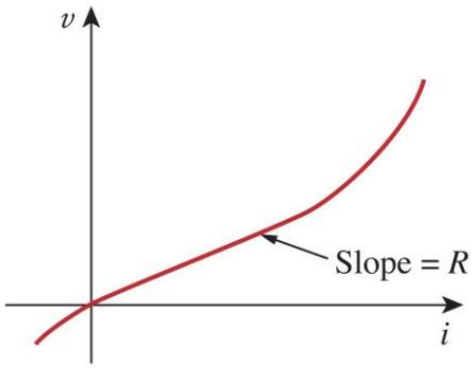
Linearity

- Not all materials obey Ohm's Law.
- Resistors that do are called linear resistors because their current voltage relationship is always linearly proportional.
- Diodes and light bulbs are examples of non-linear elements.

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(a)



(b)

Power Dissipation

- Running current through a resistor **dissipates power**.

$$p = vi = i^2 R = \frac{v^2}{R}$$

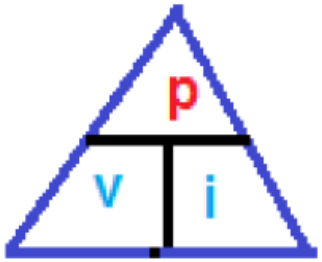
- The power dissipated is a non-linear function of current or voltage.
- Power dissipated is always **positive**.
- A resistor can never generate power.

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The power dissipated by a resistor can be expressed in terms of R.

$$p = vi = (iR)i = i^2R$$

$$= vi = v\left(\frac{v}{R}\right) = \frac{v^2}{R}$$



A resistor always absorbs power from the circuit.

A battery supplies power

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Example 2.1

An electric iron draws 2 A at 120 V. Find its resistance.

Solution:

From Ohm's law,

$$R = \frac{v}{i} = \frac{120}{2} = 60\ \Omega$$

Practice Problem 2.1

The essential component of a toaster is an electrical element (a resistor) that converts electrical energy to heat energy. How much current is drawn by a toaster with resistance 15 Ω at 110 V?

Answer: 7.333 A.

P.P.2.1 $i = V/R = 110/15 =$ **7.333 A**

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Example 2.2

In the circuit shown in Fig. 2.8, calculate the current i, the conductance G, and the power p.

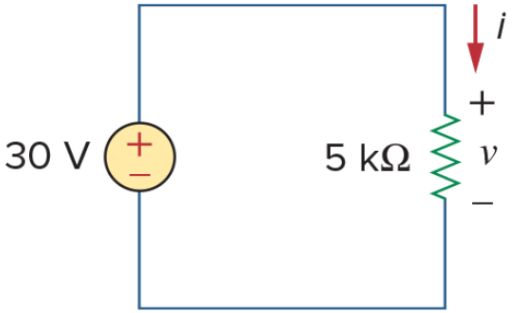


Figure 2.8
For Example 2.2.

Solution:

The voltage across the resistor is the same as the source voltage (30 V) because the resistor and the voltage source are connected to the same pair of terminals. Hence, the current is

$$i = \frac{v}{R} = \frac{30}{5 \times 10^3} = 6 \text{ mA}$$

The conductance is

$$G = \frac{1}{R} = \frac{1}{5 \times 10^3} = 0.2 \text{ mS}$$

We can calculate the power in various ways using either Eqs. (1.7), (2.10), or (2.11).

$$p = vi = 30\left(6 \times 10^{-3}\right) = 180 \text{ mW}$$

or

$$p = i^2 R = \left(6 \times 10^{-3}\right)^2 5 \times 10^3 = 180 \text{ mW}$$

or

$$p = v^2 G = (30)^2 0.2 \times 10^{-3} = 180 \text{ mW}$$

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Practice Problem 2.2

For the circuit shown in Fig. 2.9, calculate the voltage v, the conductance G, and the power p.

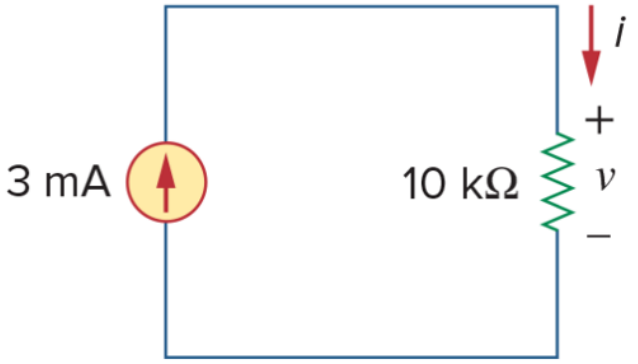


Figure 2.9
For Practice Prob. 2.2

P.P.2.2

(a) $\mathbf{v = iR} = 3\ \text{mA}*[10\ \text{kohms}] = \mathbf{30\ V}$

(b) $\mathbf{G = 1/R} = 1/10\ \text{kohms} = \mathbf{100\ \mu S}$

(c) $\mathbf{p = vi} = 30\ \text{volts}*[3\ \text{mA}] = \mathbf{90\ mW}$

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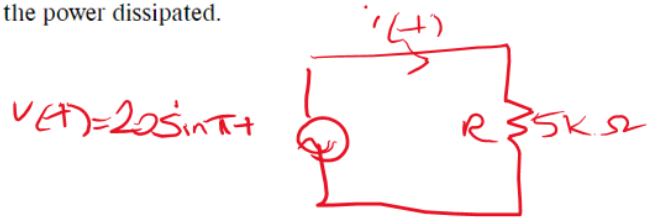
Example 2.3

A voltage source of 20 sin πt V is connected across a 5-kΩ resistor. Find the current through the resistor and the power dissipated.

Solution:

$V = IR$

$$i = \frac{v}{R} = \frac{20 \sin \pi t}{5 \times 10^3} = 4 \sin \pi t \text{ mA}$$



Hence,

$20 \sin \pi t \times 4 \sin \pi t$

$$p = vi = 80 \sin^2 \pi t \text{ mW}$$

Practice Problem 2.3

A resistor absorbs an instantaneous power of 30 $\cos^2 t$ mW when connected to a voltage source $v = 15 \cos t$ V. Find i and R.

Answer: 2 cos t mA, 7.5 kΩ.

P.P.2.3

p = vi which leads to **i = p/v** = [30 $\cos^2$ (t) mW]/[15cos(t) V]

or i = **2cos(t) mA**

R = v/i = 15cos(t)V/2cos(t)mA = **7.5 kΩ**

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Nodes Branches and Loops

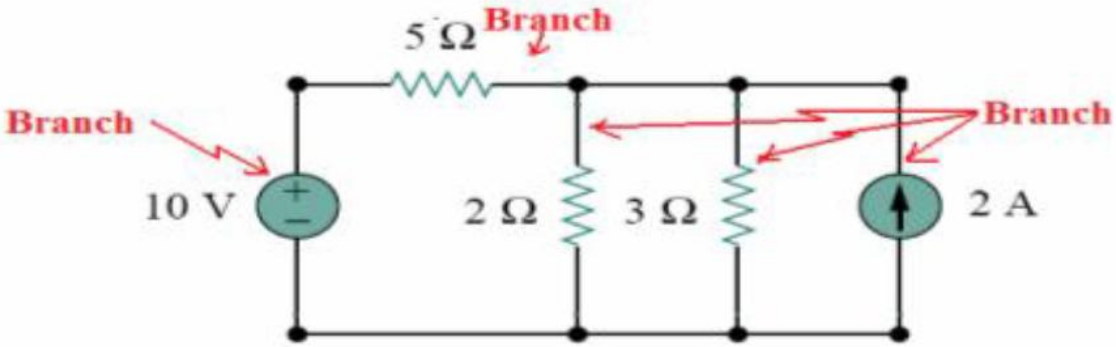
- Circuit elements can be interconnected in multiple ways.
- To understand this, we need to be familiar with some network topology concepts.
- A **branch** represents a single element such as a voltage source or a resistor.
- A **node** is the point of connection between two or more branches.
- A **loop** is any closed path in a circuit.

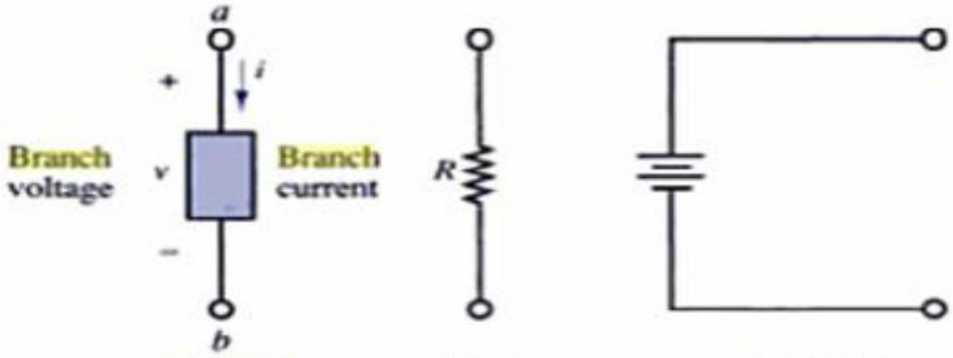
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BRANCHES

A branch represents a single element such as a voltage source or a resistor.

In other words, a branch represents any two-terminal element.



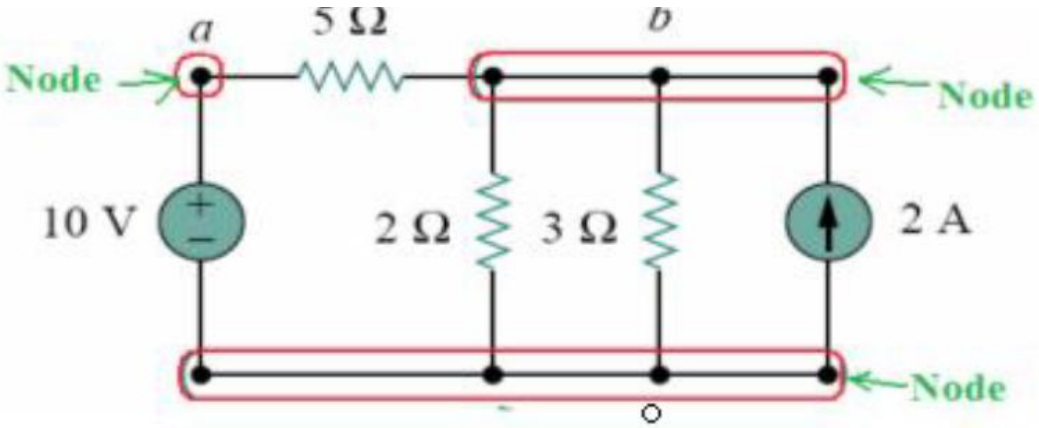


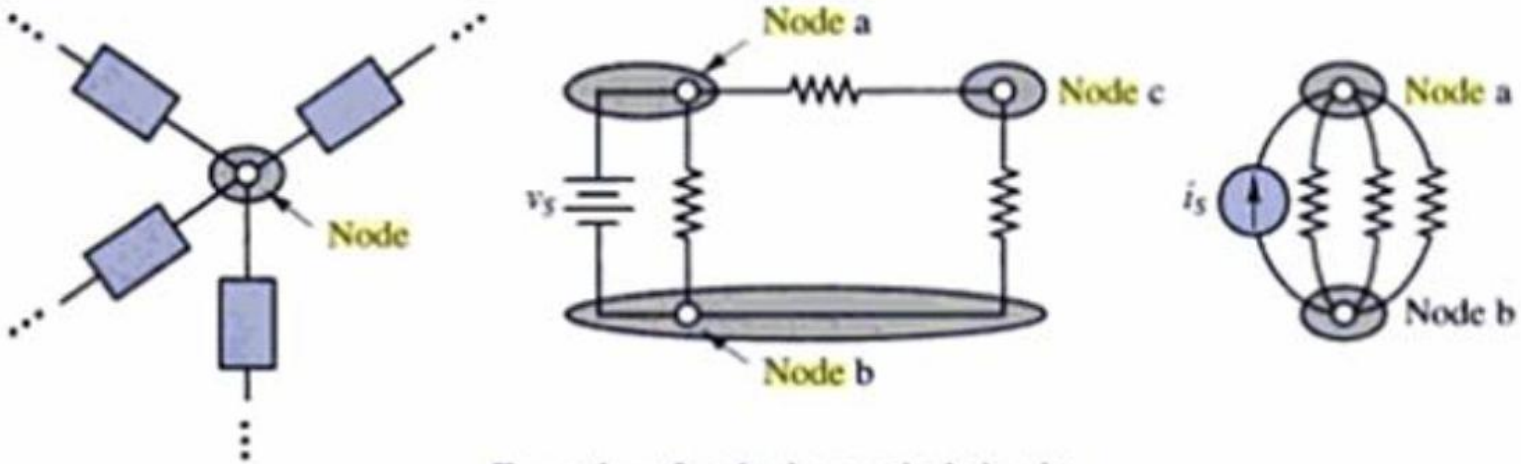
Examples of circuit branches

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NODES

A node is the point of connection between two or more branches.

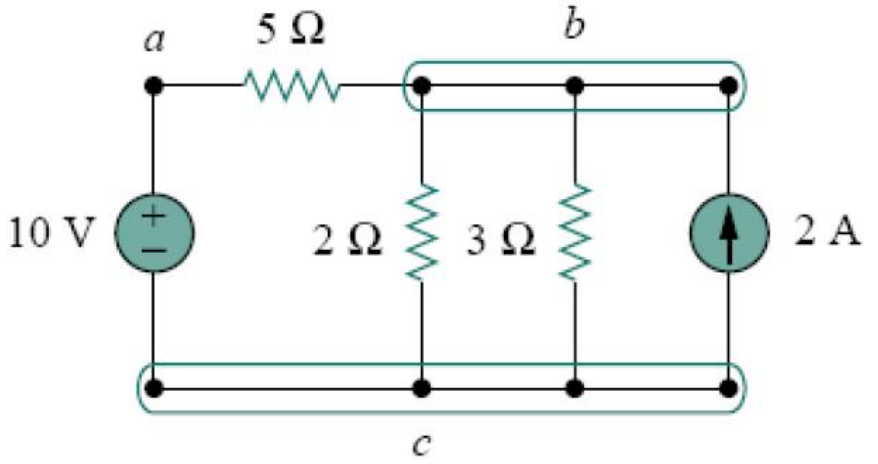


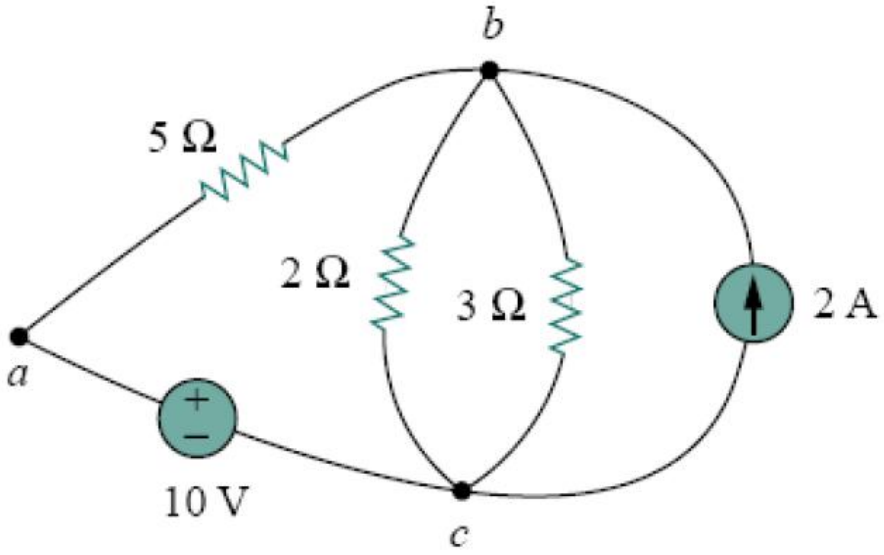


Examples of nodes in practical circuits

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(A circuit can be drawn in several ways)

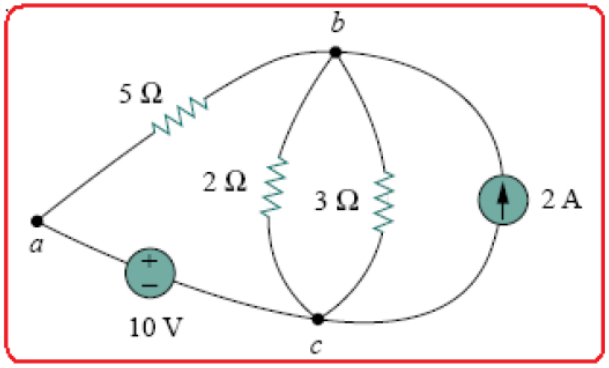


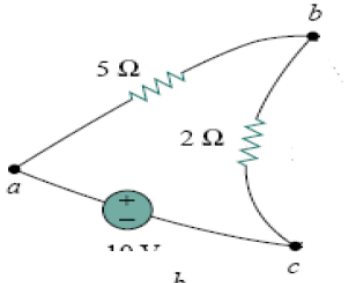


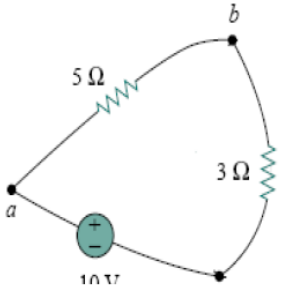
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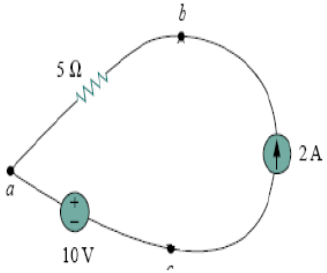
A loop is any closed path in a circuit.

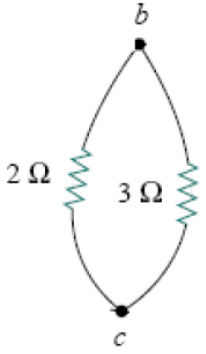
A loop is a closed path formed by starting at a node, passing through a set of nodes, and returning to the starting node without passing through any node more than once.

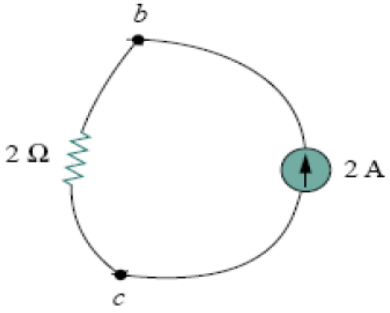


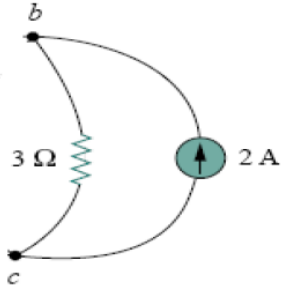








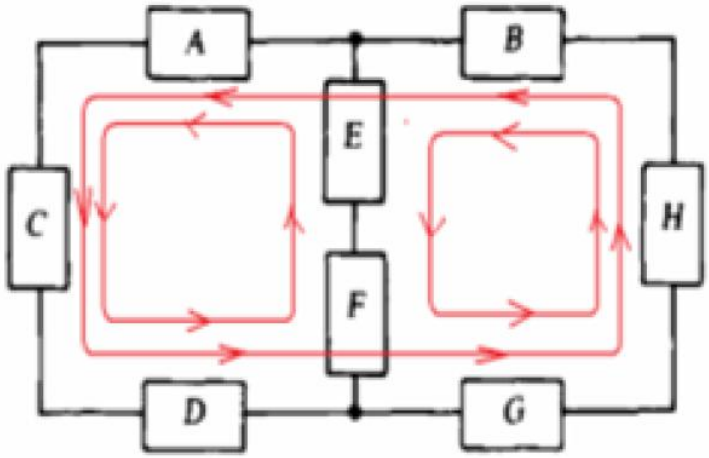




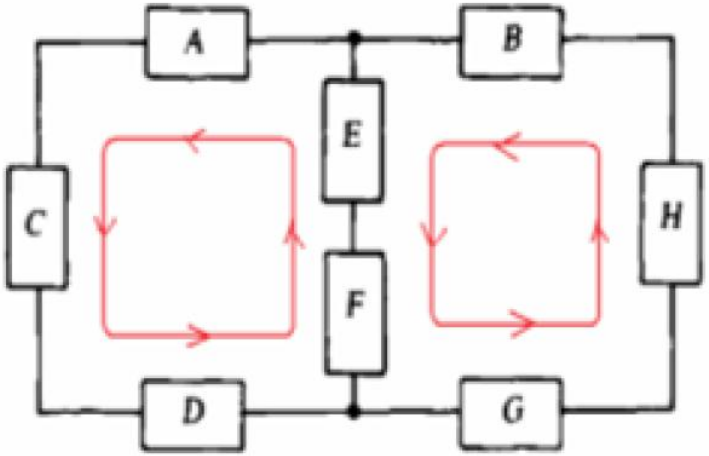
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A *mesh* is a loop that does not have a closed path in its interior.

No components are inside a mesh.



LOOP



Mesh

Relation between Nodes ,Branch and Loop

$$b = l + n - 1$$

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Example 2.4

Determine the number of branches and nodes in the circuit shown in Fig. 2.12. Identify which elements are in series and which are in parallel.

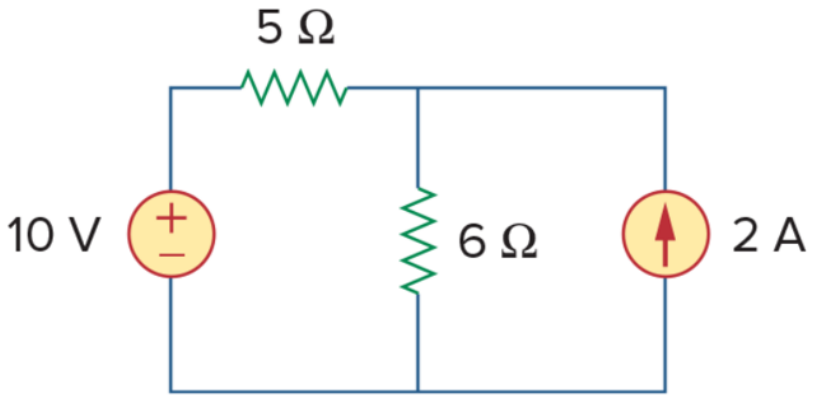


Figure 2.12
For Example 2.4.

Solution:

Since there are four elements in the circuit, the circuit has four branches: 10 V, 5 Ω, 6 Ω, and 2 A. The circuit has three nodes as identified in Fig. 2.13. The 5-Ω resistor is in series with the 10-V voltage source because the same current would flow in both. The 6-Ω resistor is in parallel with the 2-A current source because both are connected to the same nodes 2 and 3.

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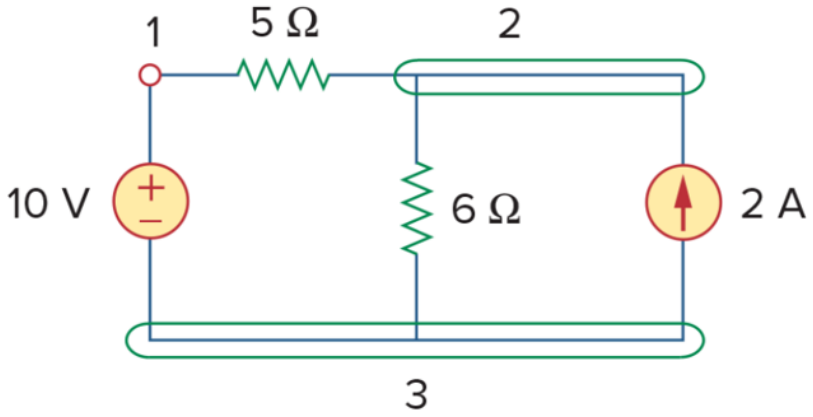


Figure 2.13

The three nodes in the circuit of Fig. 2.12.

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Practice Problem 2.4

How many branches and nodes does the circuit in Fig. 2.14 have? Identify the elements that are in series and in parallel.

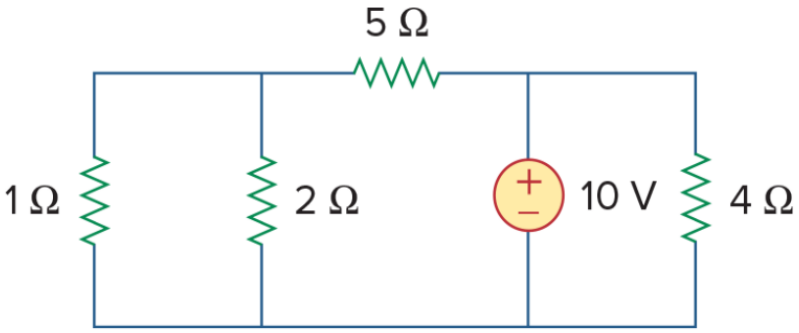


Figure 2.14
For Practice Prob. 2.4.

Answer: Five branches and three nodes are identified in Fig. 2.15. The 1-Ω and 2-Ω resistors are in parallel. The 4-Ω resistor and 10-V source are also in parallel.

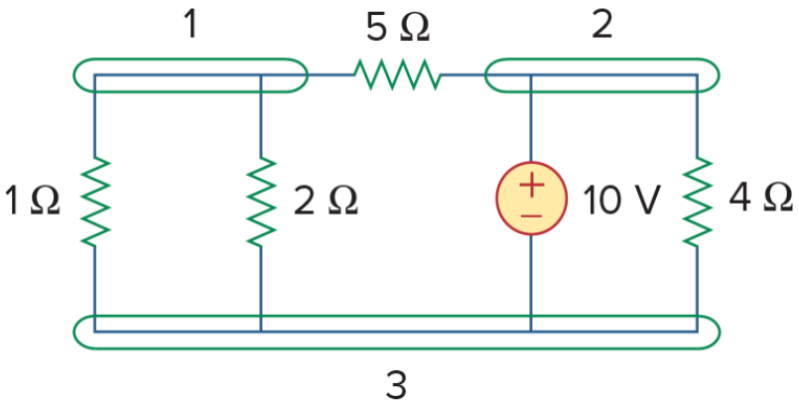


Figure 2.15
Answer for Practice Prob. 2.4.

5 branches and **3** nodes. The 1 ohm and 2 ohm resistors are in parallel. The 4 ohm resistor and the 10 volt source are also in parallel.

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Network Topology

- A **loop** is independent if it contains at least one branch not shared by any other independent loops.
- Two or more elements are in **series** if they share a **single node** and thus carry the same current.
- Two or more elements are in **parallel** if they are connected to the **same two nodes** and thus have the same voltage.

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Kirchhoff's Laws

Ohm’s law is not sufficient for circuit analysis.

Kirchhoff's laws complete the needed tools.

There are two laws:

- Kirchhoff's **Current law (KCL).**
- Kirchhoff's **Voltage law (KVL).**

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KCL

- Kirchhoff's current law is based on conservation of charge.
- It states that the algebraic sum of currents entering a node (or a closed boundary) is **zero**.
- It can be expressed as:

$$\sum_{n=1}^{N} i_n = 0$$

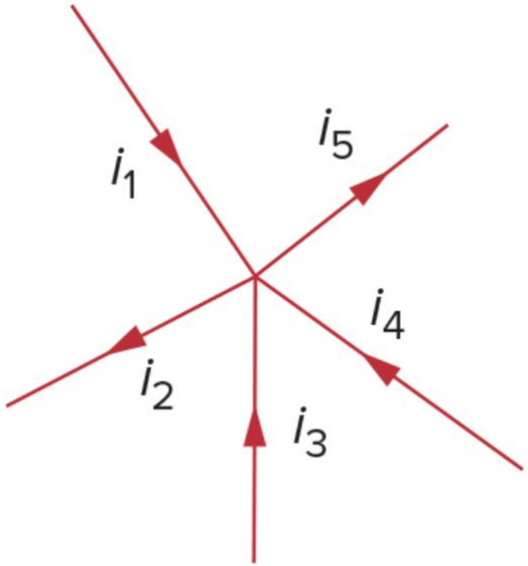
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KCL

Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

Mathematically, KCL implies that

$$\sum_{n=1}^{N} i_n = 0$$



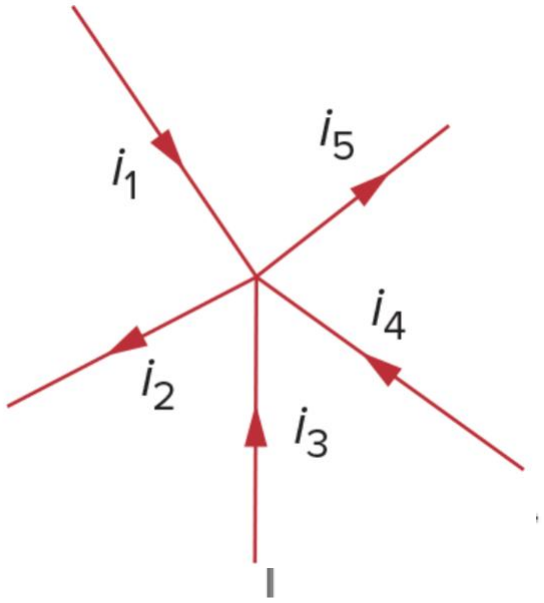
Applying KCL gives $i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$

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KCL

Alternative form of KCL:

The sum of the currents entering a node is equal to the sum of the currents leaving the node.



$$i_1 + i_3 + i_4 = i_2 + i_5$$

(current entering) (current leaving)

© McGraw Hill

KVL

- Kirchhoff's voltage law is based on conservation of energy.
- It states that the algebraic sum of voltages around a closed path (or loop) is zero.
- It can be expressed as:

$$\sum_{m=1}^{M} v_m = 0$$

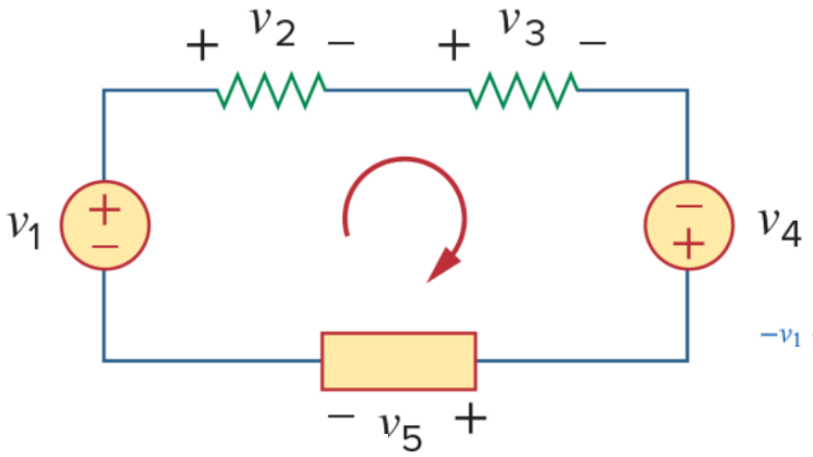
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Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

Expressed mathematically, KVL states that

$$\sum_{m=1}^{M} v_m = 0$$

To illustrate KVL, consider the circuit in Fig. 2.19.



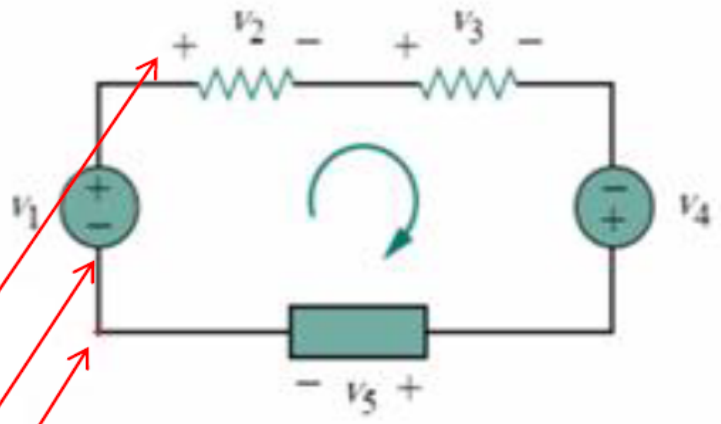
$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$

Sum of voltage drops = Sum of voltage rises

$v_2 + v_3 + v_5 = v_1 + v_4$

© McGraw Hill

Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.



We start from any one corner

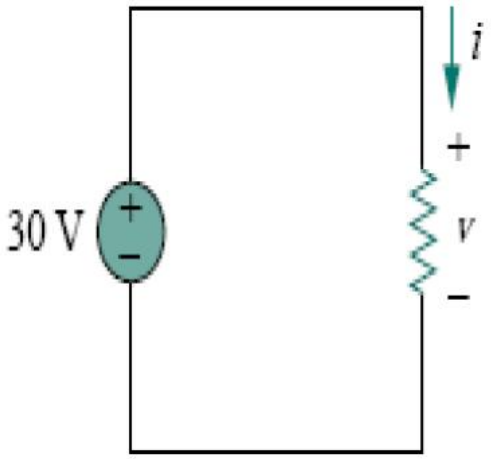
(I prefer left bottom corner, and go clockwise)

Note down the sign that comes first,
Write down that sign with voltage name

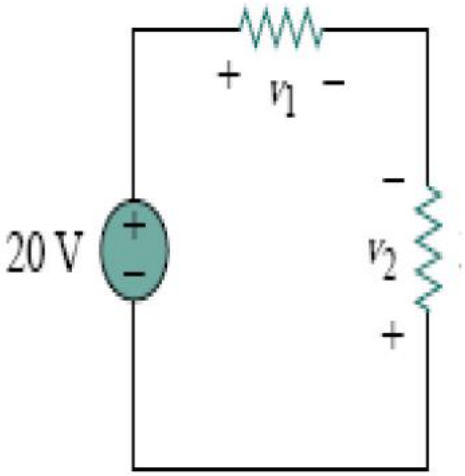
$$-v_1 + v_2 + v_3 - v_4 + v_5 = 0$$

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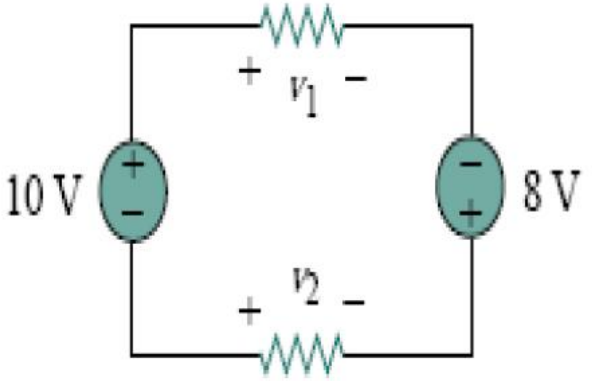
Let us do some practice of writting voltage equations.



-30 + v = 0



- 20 + v1 - v2 = 0



- 10 + v1 - 8 - v2 = 0

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Example 2.5

For the circuit in Fig. 2.21(a), find voltages v_1 and v_2.

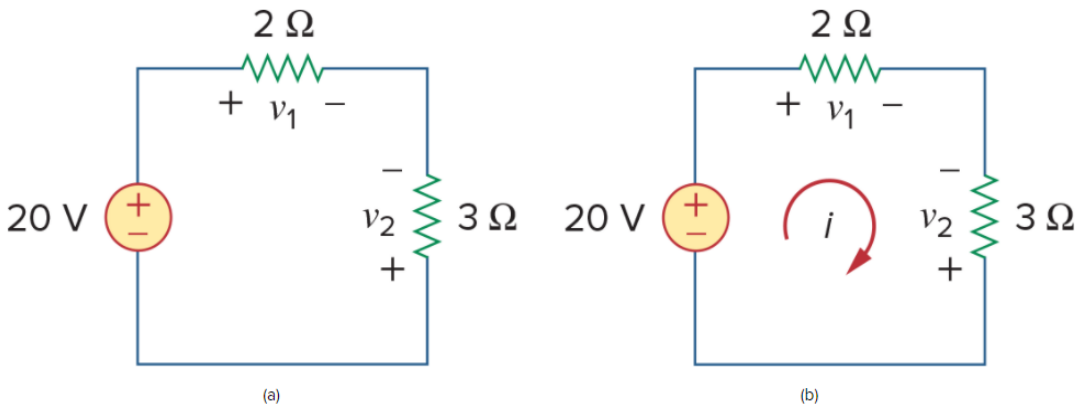


Figure 2.21
For Example 2.5.

Solution:

To find v_1 and v_2 we apply Ohm's law and Kirchhoff's voltage law. Assume that current i flows through the loop as shown in Fig. 2.21(b). From Ohm's law,

$$v_1 = 2i, \quad v_2 = -3i \tag{2.5.1}$$

Applying KVL around the loop gives

$$-20 + v_1 - v_2 = 0 \tag{2.5.2}$$

Substituting Eq. (2.5.1) into Eq. (2.5.2), we obtain

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4 \text{ A}$$

Substituting i in Eq. (2.5.1) finally gives

$$v_1 = 8 \text{ V}, \quad v_2 = -12 \text{ V}$$

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Practice Problem 2.5

Find v_1 and v_2 in the circuit of Fig. 2.22.

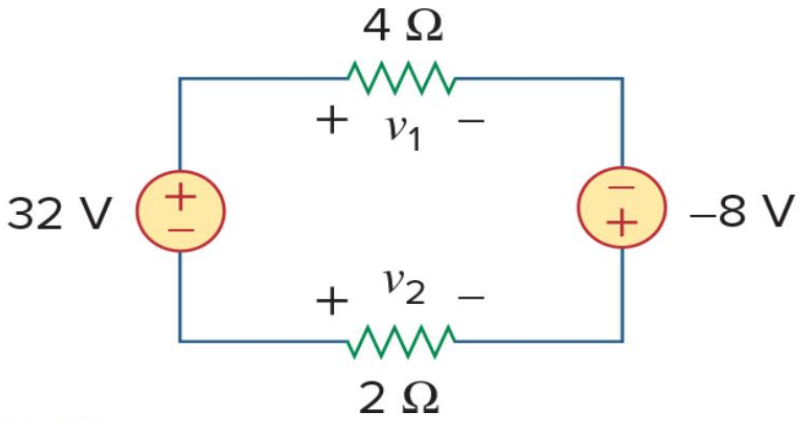


Figure 2.22
For Practice Prob. 2.5.

Answer: 16 V, –8 V.

Applying **KVL** to the loop we get:

$$-32 + 4i - (-8) + 2i = 0 \text{ which leads to } i = 24/6 = 4A$$

$$v_1 = 4i = \mathbf{16\ V} \quad \text{and} \quad v_2 = -2i = \mathbf{-8\ V}$$

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Example 2.6

Determine v_o and i in the circuit shown in Fig. 2.23(a).

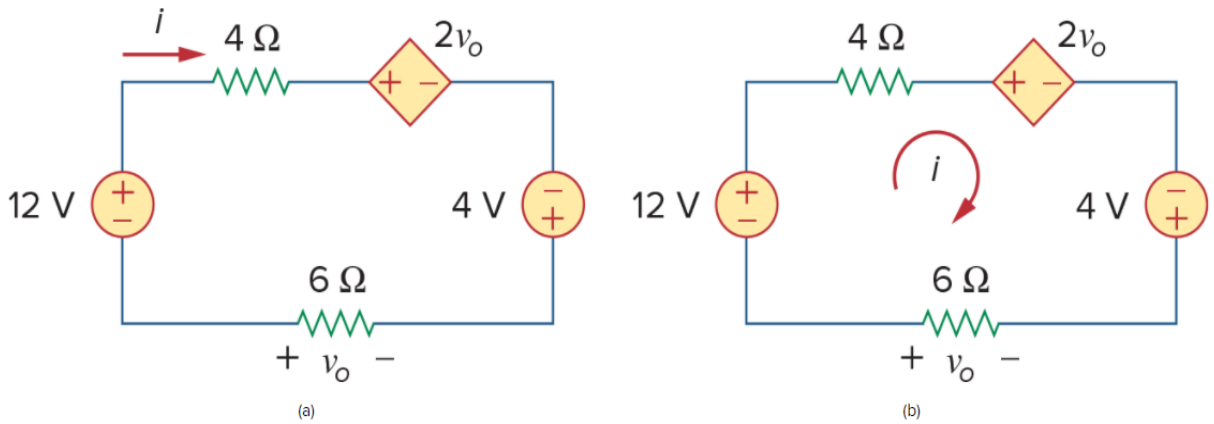


Figure 2.23
For Example 2.6.

Solution:

We apply KVL around the loop as shown in Fig. 2.23(b). The result is

$$-12 + 4i + 2v_o - 4 + 6i = 0 \tag{2.6.1}$$

Applying Ohm's law to the 6-Ω resistor gives

$$v_o = -6i \tag{2.6.2}$$

Substituting Eq. (2.6.2) into Eq. (2.6.1) yields

$$-16 + 10i - 12i = 0 \quad \Rightarrow \quad i = -8 \text{ A}$$

and $v_o = 48$ V.

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Practice Problem 2.6

Find v_x and v_o in the circuit of Fig. 2.24.

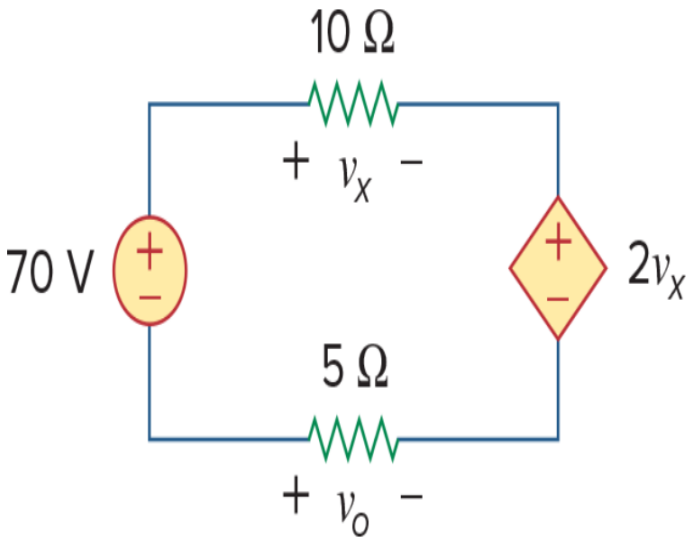


Figure 2.24
For Practice Prob. 2.6.

Answer: 20 V, –10 V.

Applying **KVL** to the loop we get:

$$-70 + 10i + 2v_x + 5i = 0$$

But, $v_x = 10i$ and $v_0 = -5i$. Hence,

$-70 + 10i + 20i + 5i = 0$ which leads to $i = 2$ A.

Thus, v_x = **20V** and v_0 = **–10 V**

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Example 2.7

Find current i_o and voltage v_o in the circuit shown in Fig. 2.25.

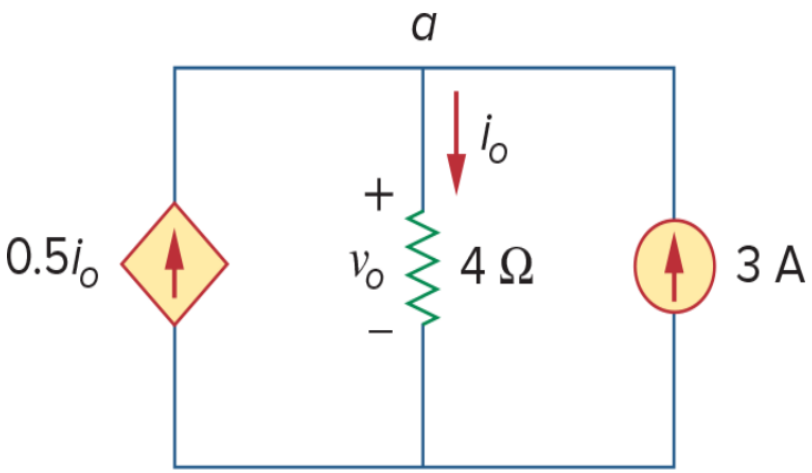


Figure 2.25
For Example 2.7.

Solution:

Applying KCL to node a, we obtain

$$3 + 0.5i_o = i_o \quad \Rightarrow \quad i_o = 6 \text{ A}$$

For the 4-Ω resistor, Ohm's law gives

$$v_o = 4i_o = 24 \text{ V}$$

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Practice Problem 2.7

Find v_o and i_o in the circuit of Fig. 2.26.

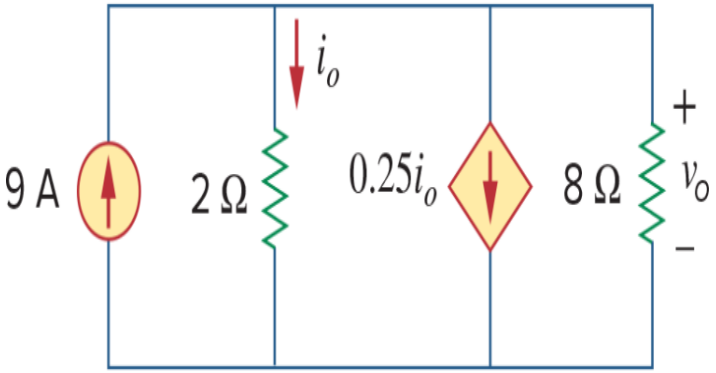


Figure 2.26
For Practice Prob. 2.7.

Answer: 12 V, 6 A.

Applying KCL, $-9 + i_0 + [i_0/4] + [v_0/8] = 0$, but $i_0 = v_0/2$

Which leads to: $9 = (v_0/2) + (v_0/8) + (v_0/8)$ thus, $v_0 =$ **12 V** and $i_0 =$ **6 A**

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Example 2.8

Find currents and voltages in the circuit shown in Fig. 2.27(a).

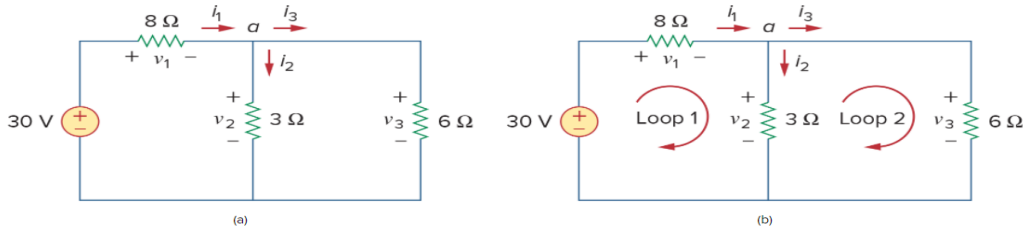


Figure 2.27
For Example 2.8.

Solution:

We apply Ohm's law and Kirchhoff's laws. By Ohm's law,

$$v_1 = 8i_1, \quad v_2 = 3i_2, \quad v_3 = 6i_3 \tag{2.8.1}$$

Since the voltage and current of each resistor are related by Ohm's law as shown, we are really looking for three things: (v_1, v_2, v_3) or (i_1, i_2, i_3). At node a, KCL gives

$$i_1 - i_2 - i_3 = 0 \tag{2.8.2}$$

Applying KVL to loop 1 as in Fig. 2.27(b),

$$-30 + v_1 + v_2 = 0$$

We express this in terms of i_1 and i_2 as in Eq. (2.8.1) to obtain

Page 43

$$-30 + 8i_1 + 3i_2 = 0$$

or

$$i_1 = \frac{(30 - 3i_2)}{8} \tag{2.8.3}$$

Applying KVL to loop 2,

$$-v_2 + v_3 = 0 \quad \Rightarrow \quad v_3 = v_2 \tag{2.8.4}$$

as expected since the two resistors are in parallel. We express v_1 and v_2 in terms of i_1 and i_2 as in Eq. (2.8.1). Equation (2.8.4) becomes

$$6i_3 = 3i_2 \quad \Rightarrow \quad i_3 = \frac{i_2}{2} \tag{2.8.5}$$

Substituting Eqs. (2.8.3) and (2.8.5) into (2.8.2) gives

$$\frac{30 - 3i_2}{8} - i_2 - \frac{i_2}{2} = 0$$

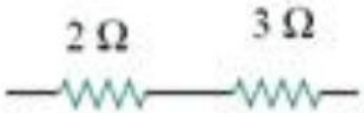
or $i_2 = 2$ A. From the value of i_2, we now use Eqs. (2.8.1) to (2.8.5) to obtain

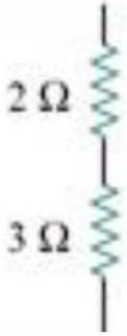
$$i_1 = 3 \text{ A}, \quad i_3 = 1 \text{ A}, \quad v_1 = 24 \text{ V}, \quad v_2 = 6 \text{ V}, \quad v_3 = 6 \text{ V}$$

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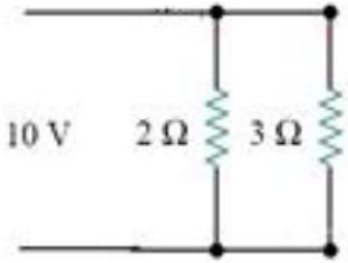
Series, Parallel Connection

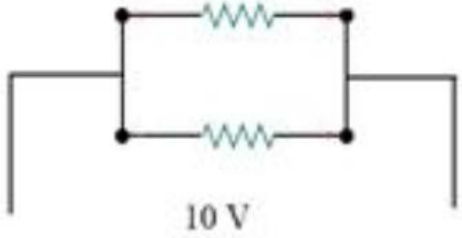
Two or more elements are in series if they are cascaded or connected sequentially and consequently carry the same current.





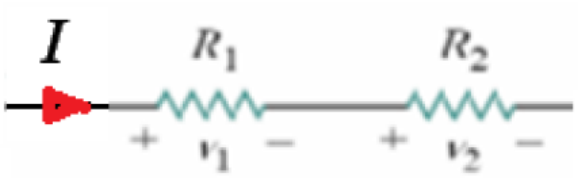
Two or more elements are in parallel if they are connected to the same two nodes and consequently have the same voltage across them.

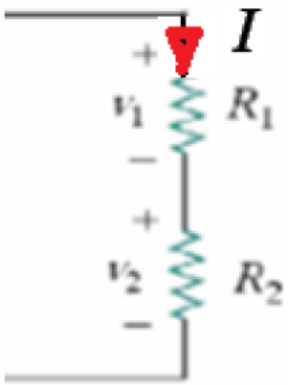




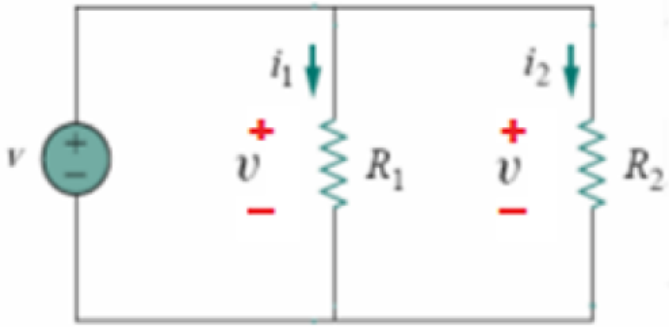
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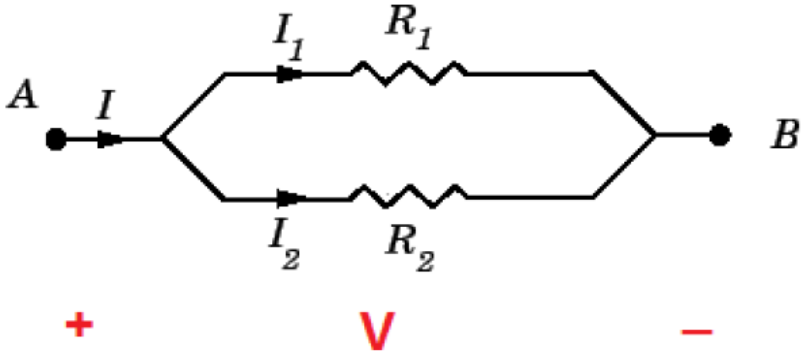
If same current flows through the resistors; the resistors are said to be in series.





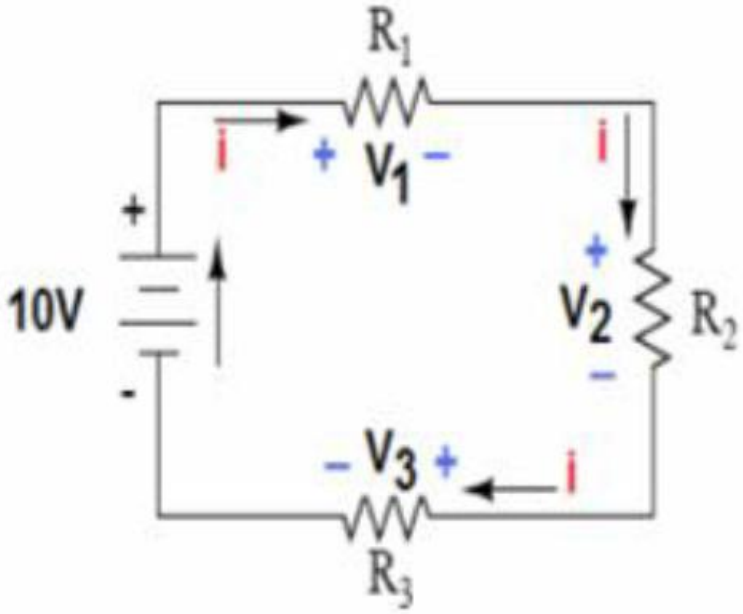
If same voltage appears across resistors, the resistors are in parallel.





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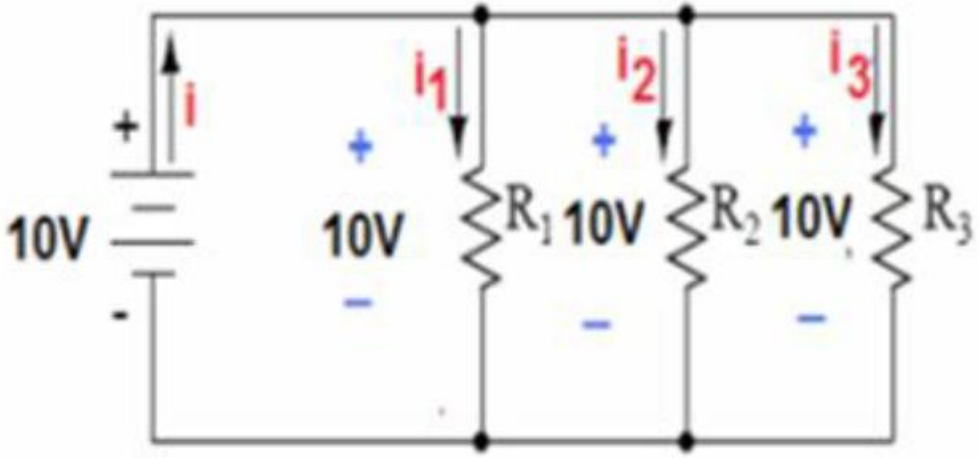
Series



Same current

Different voltage

Parallel



Different current

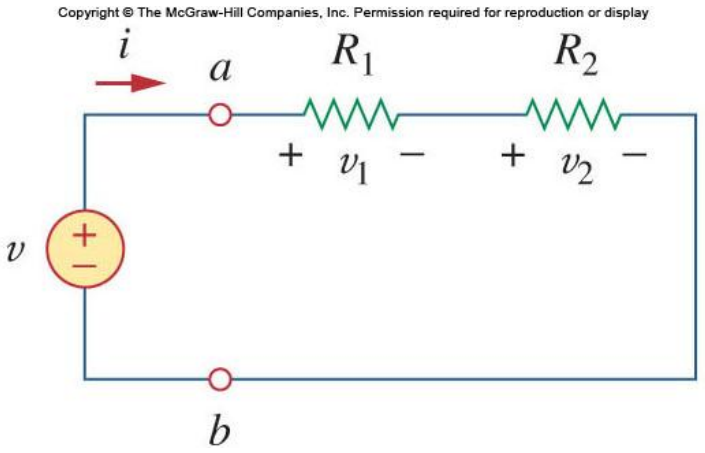
Same voltage

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Series Resistors

- Two resistors are considered in **series** if the **same current** pass through them.
- Take the circuit shown:
- Applying Ohm's law to both resistors.

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$$v_1 = iR_1 \quad v_2 = iR_2$$

- If we apply KVL to the loop we have:

$$-v + v_1 + v_2 = 0$$

Series Resistors 2

- Combining the two equations:

$$v = v_1 + v_2 = i\left(R_1 + R_2\right)$$

- From this we can see there is an equivalent resistance of the two resistors:

$$R_{eq} = R_1 + R_2$$

- For N resistors in series:

$$R_{eq} = \sum_{n=1}^{N} R_n$$

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The equivalent resistance of any number of resistors connected in series is the sum of the individual resistances.

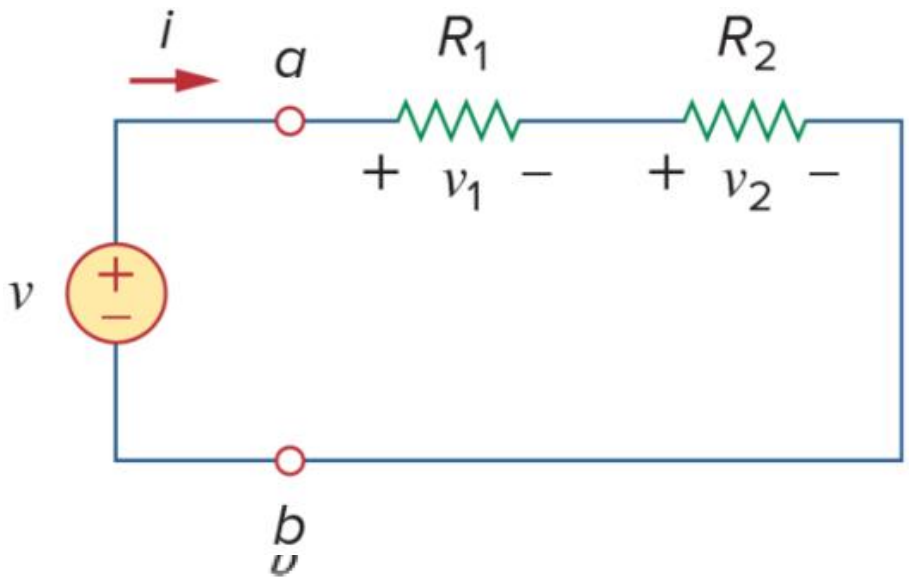
For N resistors in series then,

$$R_{eq} = R_1 + R_2 + \cdots + R_N \tag{2.30}$$

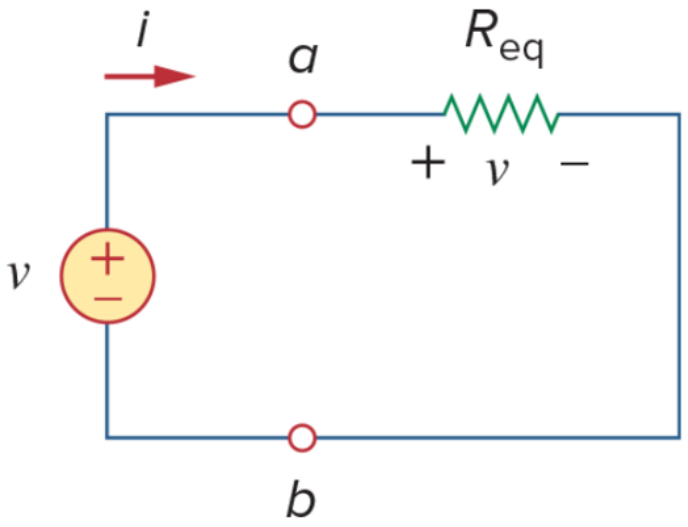
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SERIES RESISTORS

$$R_{eq} = R_1 + R_2$$



A single-loop circuit with two resistors in series.



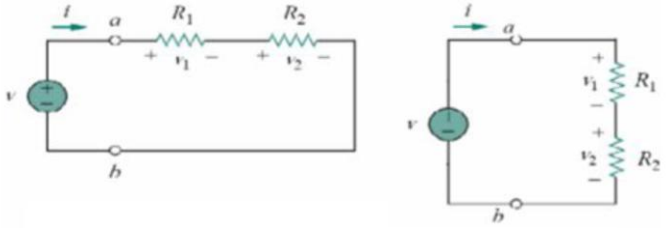
Equivalent circuit

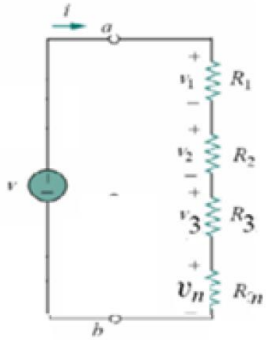
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Voltage Division

- The voltage drop across any one resistor can be known.
- The current through all the resistors is the same, so using Ohm's law:

$$v_1 = \frac{R_1}{R_1 + R_2} v \qquad v_2 = \frac{R_2}{R_1 + R_2} v$$





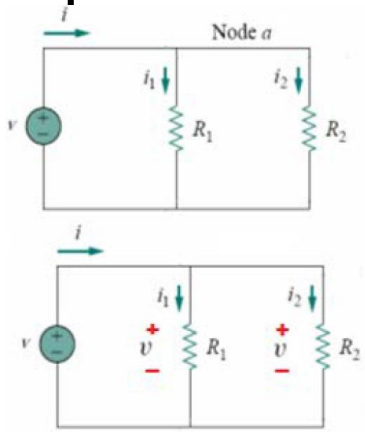
- This is the principle of voltage division.

$$v_n = \frac{R_n}{R_1 + R_2 + \cdots + R_N} \times v$$

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Parallel Resistors

- When resistors are in parallel, the voltage drop across them is the same. $v = i_1 R_1 = i_2 R_2$



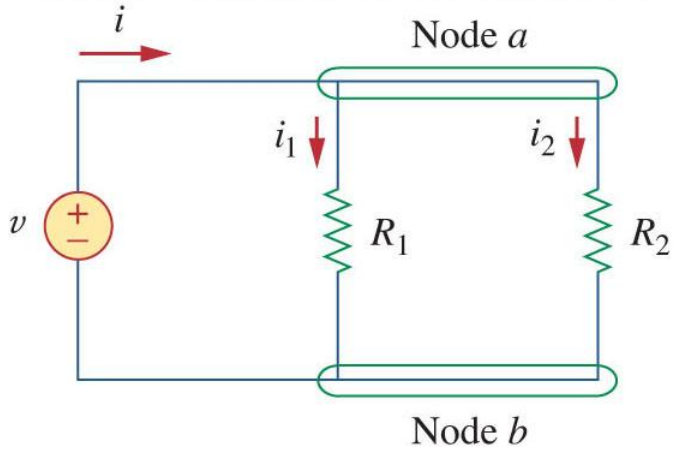
- By KCL, the current at node a is

$$i = i_1 + i_2$$

- The equivalent resistance is:

$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

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The equivalent resistance of two parallel resistors is equal to the product of their resistances divided by their sum.

3 resistors in parallel $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$

N resistors in parallel $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N}$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \cdots + \frac{1}{R_N} \tag{2.38}$$

It is often more convenient to use conductance rather than resistance when dealing with resistors in parallel. From Eq. (2.38), the equivalent conductance for N resistors in parallel is

$$G_{eq} = G_1 + G_2 + G_3 + \cdots + G_N \tag{2.40}$$

where $G_{eq} = 1/R_{eq}, G_1 = 1/R_1, G_2 = 1/R_2, G_3 = 1/R_3, \ldots, G_N = 1/R_N$.

Conductances in parallel behave as a single conductance whose value is equal to the sum of the individual conductances.

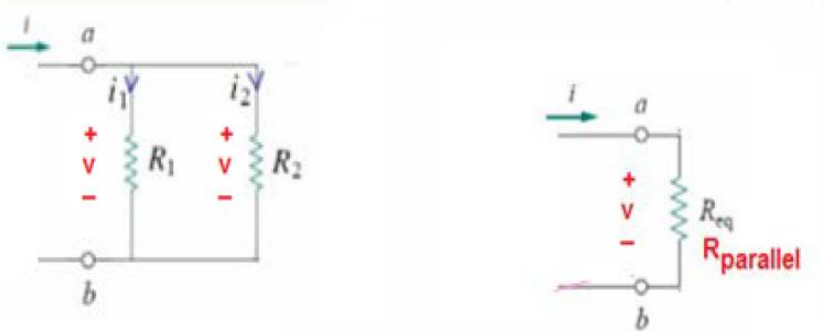
The equivalent conductance of resistors connected in parallel is the sum of their individual conductances.

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Current Division

- Given the current entering the node, the voltage drop across the equivalent resistance will be the same as that for the individual resistors.

$$v = iR_{eq} = \frac{iR_1R_2}{R_1 + R_2}$$



- This can be used in combination with Ohm's law to get the current through each resistor:

$$i_1 = \frac{iR_2}{R_1 + R_2} \qquad i_2 = \frac{iR_1}{R_1 + R_2}$$

$$i_1 = \frac{R_2}{R_1 + R_2} \times i \quad , \quad i_2 = \frac{R_1}{R_1 + R_2} \times i$$

(note the mismatch at the numerator)

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Example 2.9

Find R_{eq} for the circuit shown in Fig. 2.34.

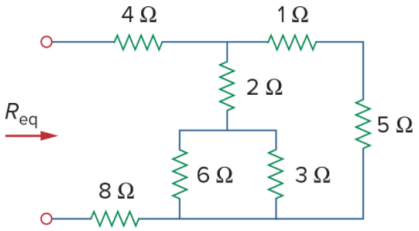


Figure 2.34
For Example 2.9.

Solution:

To get R_{eq}, we combine resistors in series and in parallel. The 6-Ω and 3-Ω resistors are in parallel, so their equivalent resistance is

$$6\ \Omega \parallel 3\Omega = \frac{6 \times 3}{6+3} = 2\ \Omega$$

(The symbol ∥ is used to indicate a parallel combination.) Also, the 1-Ω and 5-Ω resistors are in series; hence their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega$$

Thus the circuit in Fig. 2.34 is reduced to that in Fig. 2.35(a). In Fig. 2.35(a), we notice that the two 2-Ω resistors are in series, so the equivalent resistance is

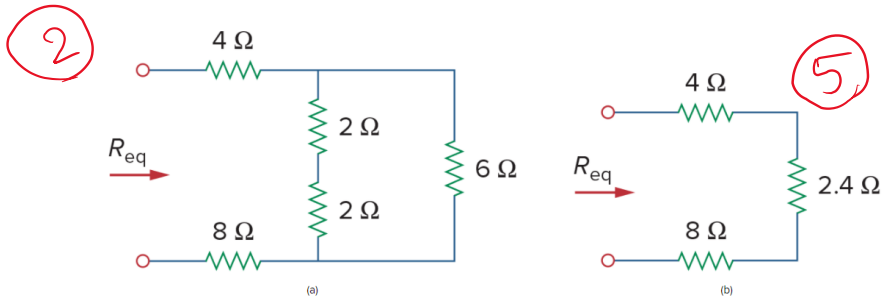


Figure 2.35
Equivalent circuits for Example 2.9.

This 4-Ω resistor is now in parallel with the 6-Ω resistor in Fig. 2.35(a); their equivalent resistance is

$$4\ \Omega \parallel 6\ \Omega = \frac{4 \times 6}{4+6} = 2.4\ \Omega$$

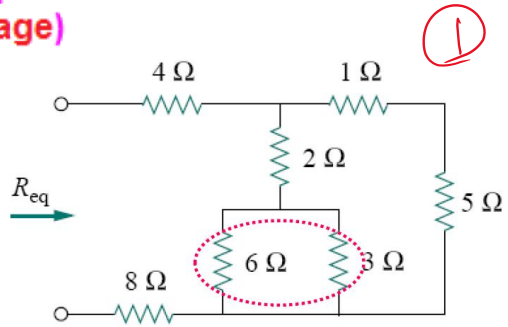
The circuit in Fig. 2.35(a) is now replaced with that in Fig. 2.35(b). In Fig. 2.35(b), the three resistors are in series. Hence, the equivalent resistance for the circuit is

$$R_{eq} = 4\ \Omega + 2.4\ \Omega + 8\ \Omega = 14.4\ \Omega$$

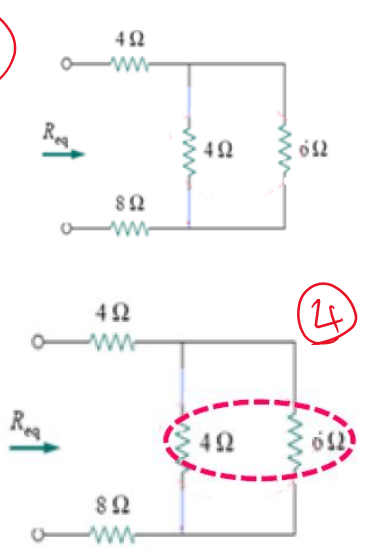
Hint:

1. Solve for the parallel components first.
2. Solve starting from Right .

Identify the branches that are in parallel (same voltage)



Adding the series components



Example 2.10

Calculate the equivalent resistance R_{ab} in the circuit in Fig. 2.37.

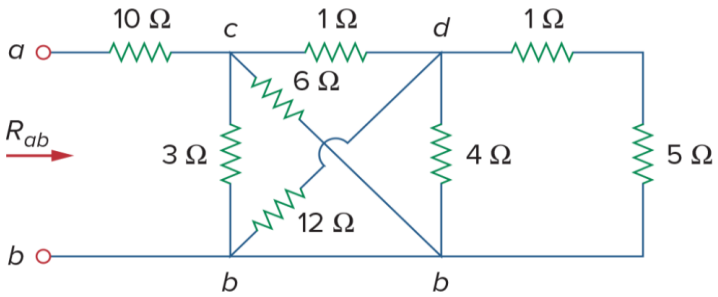
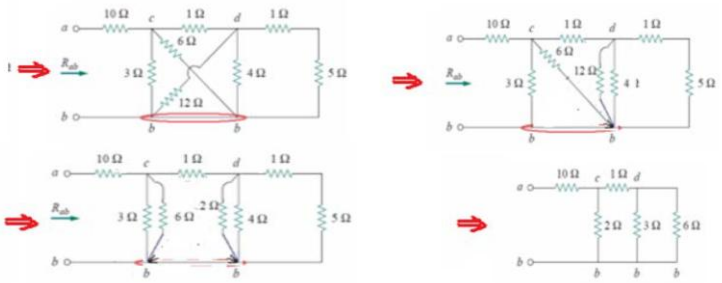


Figure 2.37
For Example 2.10.

1. Solve for the parallel components first.
2. Solve starting from Right .



Solution:

The 3-Ω and 6-Ω resistors are in parallel because they are connected to the same two nodes c and b. Their combined resistance is

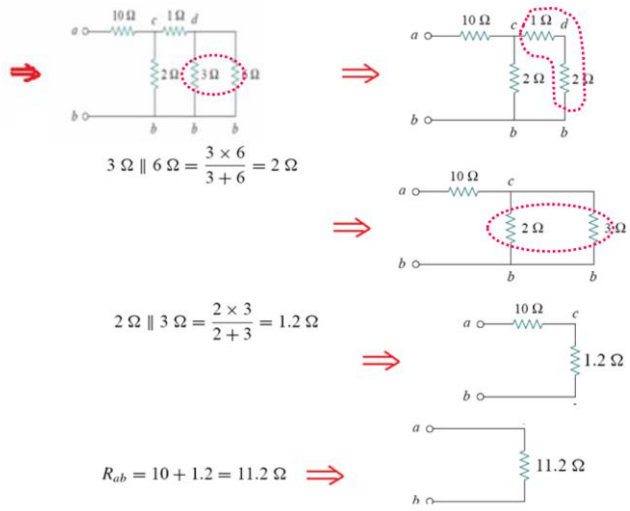
$$3\ \Omega \parallel 6\ \Omega = \frac{3 \times 6}{3+6} = 2\ \Omega \tag{2.10.1}$$

Similarly, the 12-Ω and 4-Ω resistors are in parallel since they are connected to the same two nodes d and b. Hence

$$12\ \Omega \parallel 4\ \Omega = \frac{12 \times 4}{12+4} = 3\ \Omega \tag{2.10.2}$$

Also the 1-Ω and 5-Ω resistors are in series; hence, their equivalent resistance is

$$1\ \Omega + 5\ \Omega = 6\ \Omega \tag{2.10.3}$$



With these three combinations, we can replace the circuit in Fig. 2.37 with that in Fig. 2.38(a). In Fig. 2.38(a), 3-Ω in parallel with 6-Ω gives 2-Ω, as calculated in Eq. (2.10.1). This 2-Ω equivalent resistance is now in series with the 1-Ω resistance to give a combined resistance of $1\ \Omega + 2\ \Omega = 3\ \Omega$. Thus, we replace the circuit in Fig. 2.38(a) with that in Fig. 2.38(b). In Fig. 2.38(b), we combine the 2-Ω and 3-Ω resistors in parallel to get

$$2\ \Omega \parallel 3\ \Omega = \frac{2 \times 3}{2+3} = 1.2\ \Omega$$

This 1.2-Ω resistor is in series with the 10-Ω resistor, so that

$$R_{ab} = 10 + 1.2 = 11.2\ \Omega$$

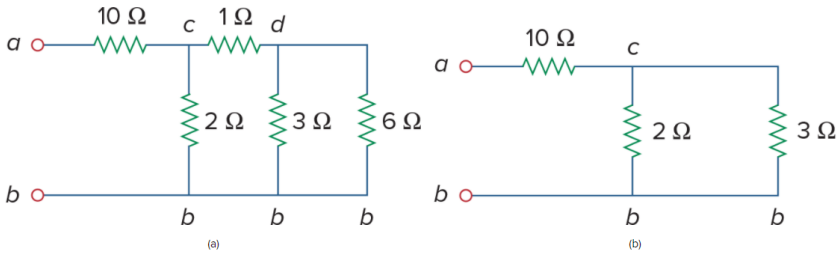


Figure 2.38
Equivalent circuits for Example 2.10.

Practice Problem 2.10

Find R_{ab} for the circuit in Fig. 2.39.

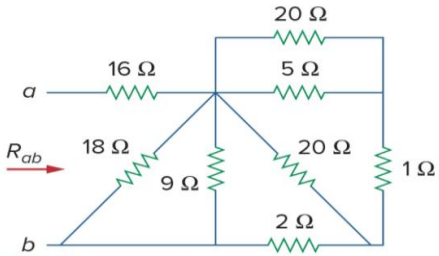
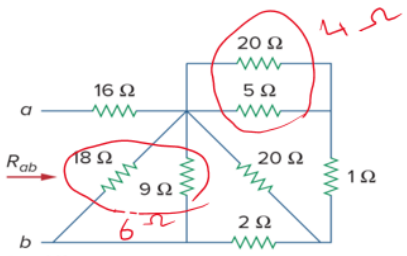


Figure 2.39
For Practice Prob. 2.10.

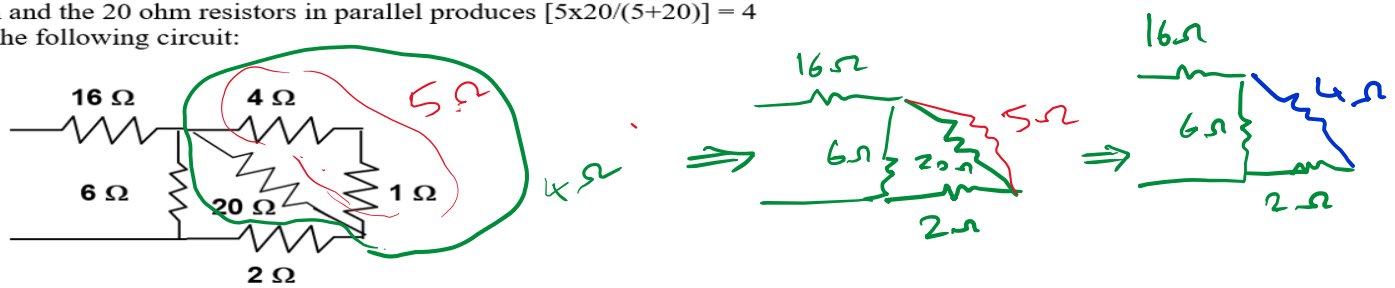


Answer: 19 Ω.

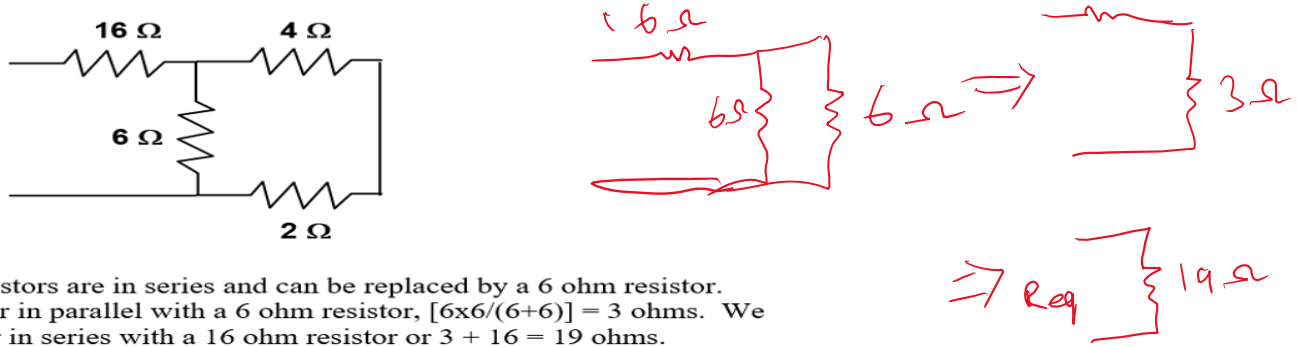
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Combining the 9-ohm resistor and the 18-ohm resistor yields [9x18]/[9+18] = 6 ohms.

Combining the 5 ohm and the 20 ohm resistors in parallel produces [5x20/(5+20)] = 4 ohms We now have the following circuit:



The 4 ohm and 1 ohm resistors can be combined into a 5 ohm resistor in parallel with a 20 ohm resistor. This will result in [5x20/(5+20)] = 4 ohms and the circuit shown below:



The 4 ohm and 2 ohm resistors are in series and can be replaced by a 6 ohm resistor. This gives a 6 ohm resistor in parallel with a 6 ohm resistor, [6x6/(6+6)] = 3 ohms. We now have a 3 ohm resistor in series with a 16 ohm resistor or 3 + 16 = 19 ohms. Therefore:

R_{eq} = 19 ohms

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Example 2.11

Find the equivalent conductance G_{eq} for the circuit in Fig. 2.40(a).

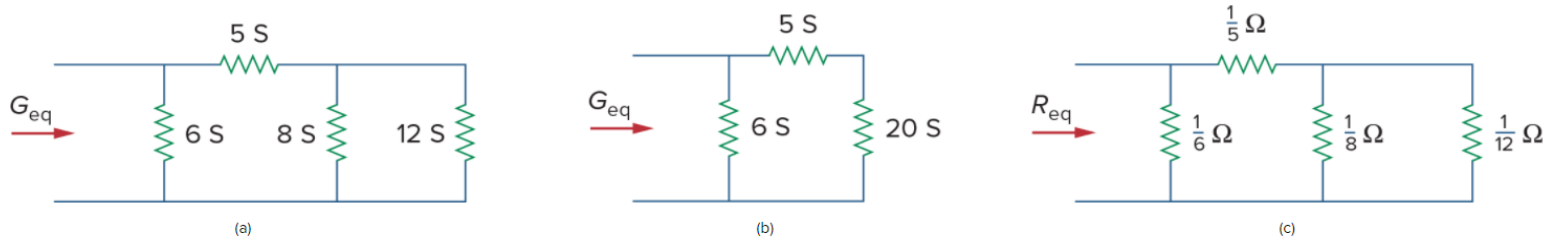


Figure 2.40
For Example 2.11: (a) original circuit, (b) its equivalent circuit, (c) same circuit as in (a) but resistors are expressed in ohms.

Solution:

The 8-S and 12-S resistors are in parallel, so their conductance is

$$8\ \text{S} + 12\ \text{S} = 20\ \text{S}$$

This 20-S resistor is now in series with 5 S as shown in Fig. 2.40(b) so that the combined conductance is

$$\frac{20 \times 5}{20 + 5} = 4\ \text{S}$$

This is in parallel with the 6-S resistor. Hence,

$$G_{eq} = 6 + 4 = 10\ \text{S}$$

We should note that the circuit in Fig. 2.40(a) is the same as that in Fig. 2.40(c). While the resistors in Fig. 2.40(a) are expressed in siemens, those in Fig. 2.40(c) are expressed in ohms. To show that the circuits are the same, we find R_{eq} for the circuit in Fig. 2.40(c).

$$R_{eq} = \frac{1}{6} \parallel \left(\frac{1}{5} + \frac{1}{8} \parallel \frac{1}{12}\right) = \frac{1}{6} \parallel \left(\frac{1}{5} + \frac{1}{20}\right) = \frac{1}{6} \parallel \frac{1}{4}$$

$$G_{eq} = \frac{1}{R_{eq}} = 10\ \text{S}$$

This is the same as we obtained previously.

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Example 2.12

Find i_o and v_o in the circuit shown in Fig. 2.42(a). Calculate the power dissipated in the 3-Ω resistor.

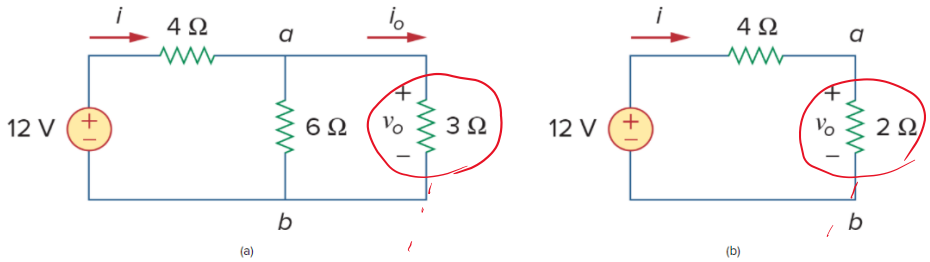


Figure 2.42
For Example 2.12: (a) original circuit, (b) its equivalent circuit.

Solution:

The 6-Ω and 3-Ω resistors are in parallel, so their combined resistance is

$$6\ \Omega \parallel 3\ \Omega = \frac{6 \times 3}{6+3} = 2\ \Omega$$

Thus, our circuit reduces to that shown in Fig. 2.42(b). Notice that v_o is not affected by the combination of the resistors because the resistors are in parallel and therefore have the same voltage v_o. From Fig. 2.42(b), we can obtain v_o in two ways. One way is to apply Ohm's law to get

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$$i = \frac{12}{4+2} = 2\ \text{A}$$

and hence, $v_o = 2i = 2 \times 2 = 4$ V. Another way is to apply voltage division, since the 12 V in Fig. 2.42(b) is divided between the 4-Ω and 2-Ω resistors. Hence,

$$v_o = \frac{2}{2+4}(12\ \text{V}) = 4\ \text{V}$$

Similarly, i_o can be obtained in two ways. One approach is to apply Ohm's law to the 3-Ω resistor in Fig. 2.42(a) now that we know v_o; thus,

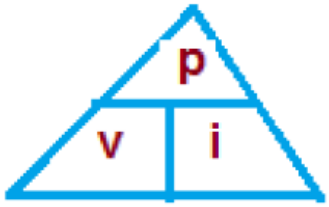
$$v_o = 3i_o = 4 \quad \Rightarrow \quad i_o = \frac{4}{3}\ \text{A}$$

Another approach is to apply current division to the circuit in Fig. 2.42(a) now that we know i, by writing

$$i_o = \frac{6}{6+3} i = \frac{2}{3}(2\ \text{A}) = \frac{4}{3}\ \text{A}$$

The power dissipated in the 3-Ω resistor is

$$p_o = v_o i_o = 4\left(\frac{4}{3}\right) = 5.333\ \text{W}$$



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Example 2.13

For the circuit shown in Fig. 2.44(a), determine: (a) the voltage v_o, (b) the power supplied by the current source, (c) the power absorbed by each resistor.

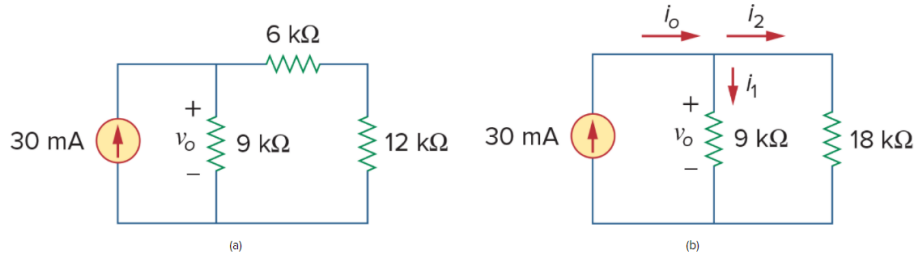


Figure 2.44
For Example 2.13: (a) original circuit, (b) its equivalent circuit.

Solution:

(a) The 6-kΩ and 12-kΩ resistors are in series so that their combined value is $6 + 12 = 18$ kΩ. Thus the circuit in Fig. 2.44(a) reduces to that shown in Fig. 2.44(b). We now apply the current division technique to find i_1 and i_2.

$$i_1 = \frac{18{,}000}{9{,}000 + 18{,}000}(30\text{ mA}) = 20\text{ mA}$$
$$i_2 = \frac{9{,}000}{9{,}000 + 18{,}000}(30\text{ mA}) = 10\text{ mA}$$

Notice that the voltage across the 9-kΩ and 18-kΩ resistors is the same, and $v_o = 9{,}000i_1 = 18{,}000i_2 = 180$ V, as expected.

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(b) Power supplied by the source is

$$p_o = v_o i_o = 180\,(30)\text{ mW} = 5.4\text{ W}$$

(c) Power absorbed by the 12-kΩ resistor is

$$p = iv = i_2\,(i_2 R) = i_2^2 R = \left(10 \times 10^{-3}\right)^2 (12{,}000) = 1.2\text{ W}$$

Power absorbed by the 6-kΩ resistor is

$$p = i_2^2 R = \left(10 \times 10^{-3}\right)^2 (6{,}000) = 0.6\text{ W}$$

Power absorbed by the 9-kΩ resistor is

$$p = \frac{v_o^2}{R} = \frac{(180)^2}{9{,}000} = 3.6\text{ W}$$

or

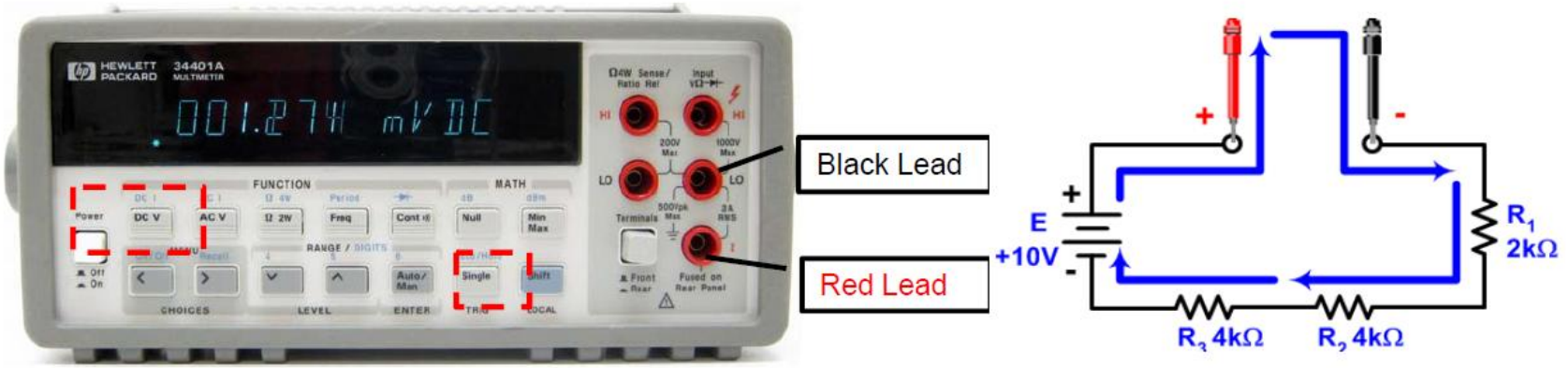
$$p = v_o i_1 = 180\,(20)\text{ mW} = 3.6\text{ W}$$

Notice that the power supplied (5.4 W) equals the power absorbed ($1.2 + 0.6 + 3.6 = 5.4$ W). This is one way of checking results.

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Measurement Devices: Ammeter

- Connect the test leads to the DMM as shown below
  - To measure current the multimeter must be connected in series with the circuit under test. Remember, current is the flow of charge. The charge must flow through the meter.
- Push the power switch to the ON position
- Press Shfit + DCI or ACI switch (depends on what you are measuring
- Connect the red and black lead to the circuit as shown in the diagram
  - Always place the red test lead at a point that is more positive than the black test lead.
- Note: the highest current range and never exceed it



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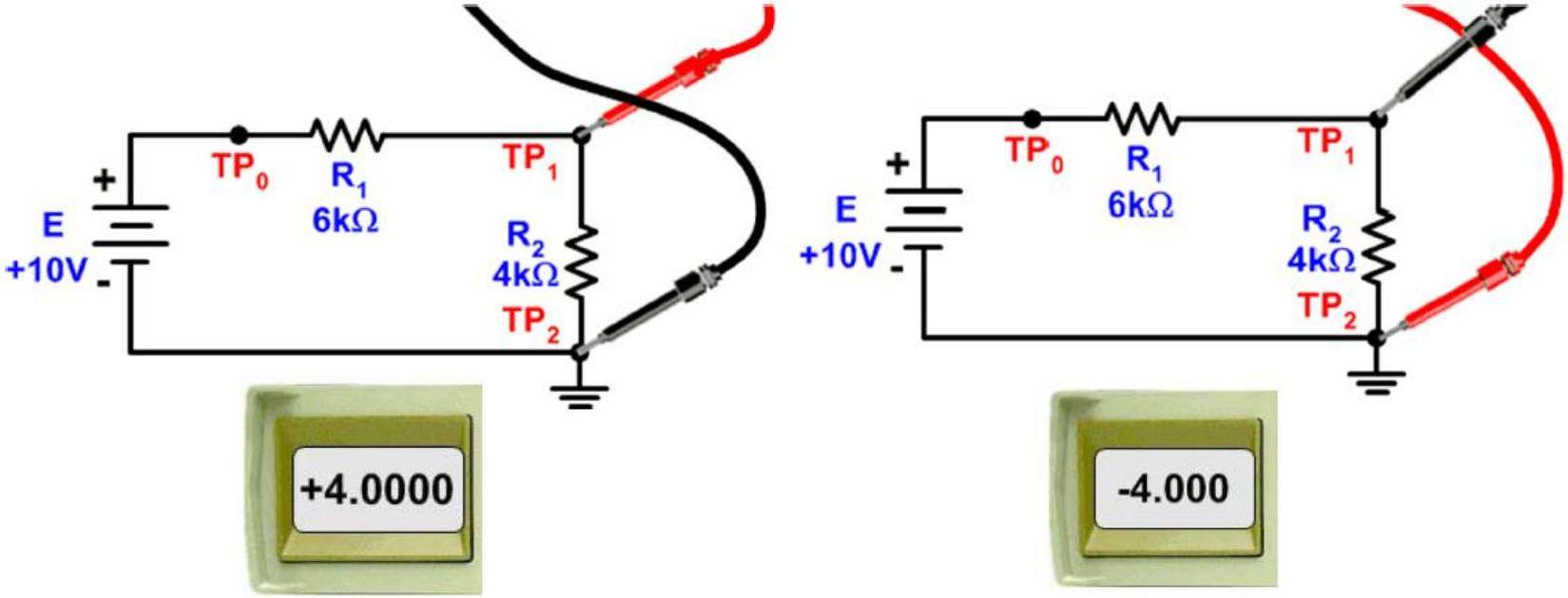
Voltmeter

- Connect the test leads to the DMM as shown below
- Push the power switch to the ON position
- Press DCV or ACV switch (depends on what you are measuring
- Connect the black lead to the circuit common and the red lead to a desired point
- Note: the highest volt range and never exceed it



Voltmeter II

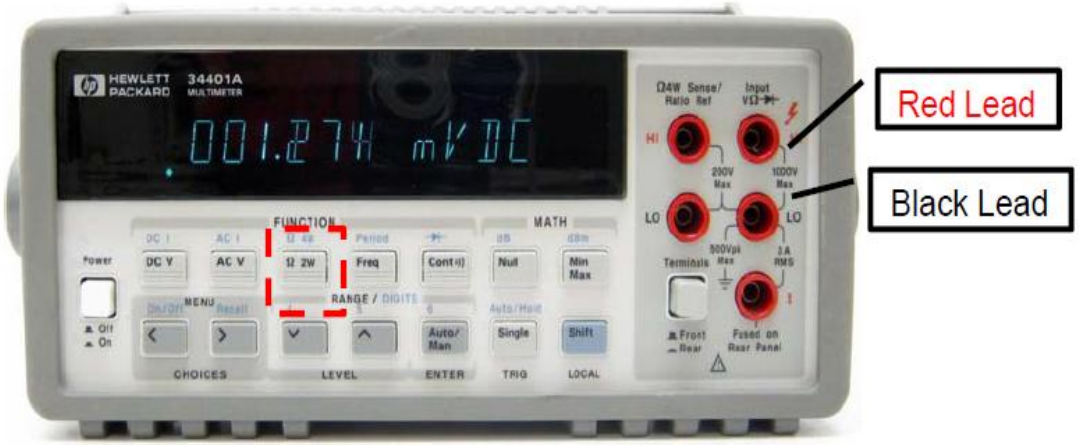
- As shown in the diagram on the right, reversing the leads (red lead on ground), will cause meter display to display a negative voltage reading.
  - If this condition should occur while measuring voltage, reverse the test leads.

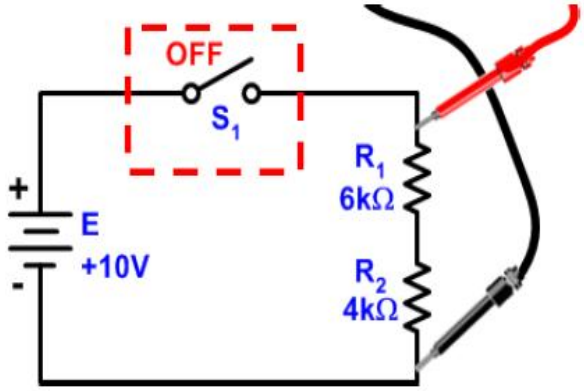


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Ohmmeter

- Connect the red and black lead to the DMM as shown in the diagram
- Push the power switch to the ON position
- Press Ω 2W switch
- Connect the test leads across the resistance being measured.
  - When taking resistance measurements always turn the power OFF! Test lead placement is not referenced to ground when measuring resistance. Placing the positive or negative lead on either side of the resistor produces the same result.

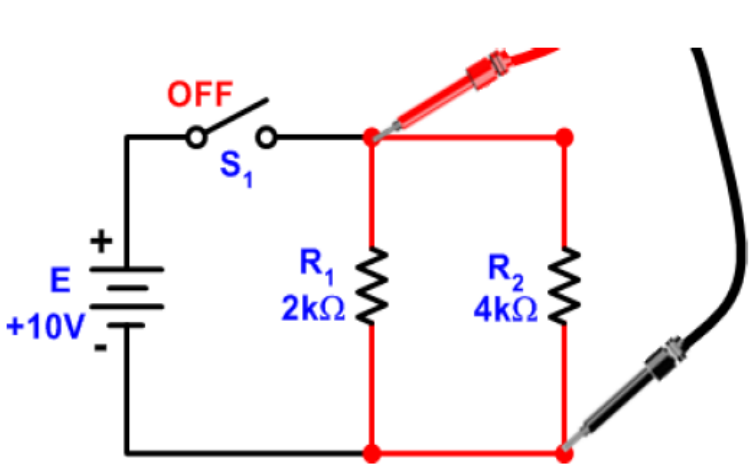


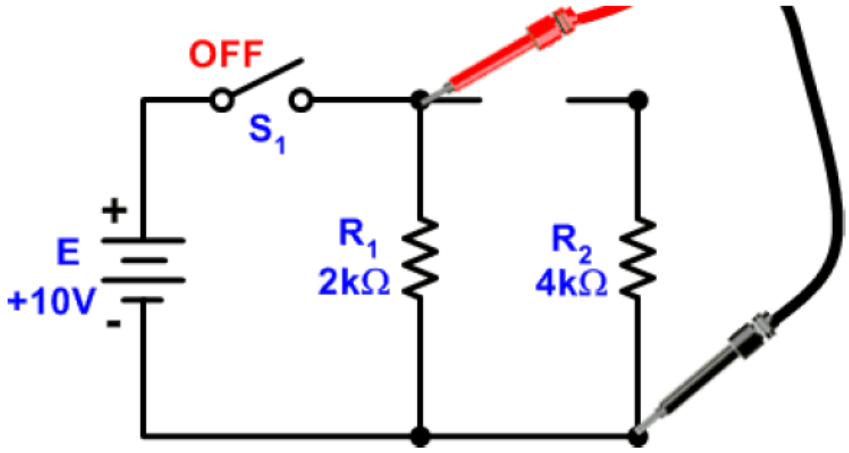


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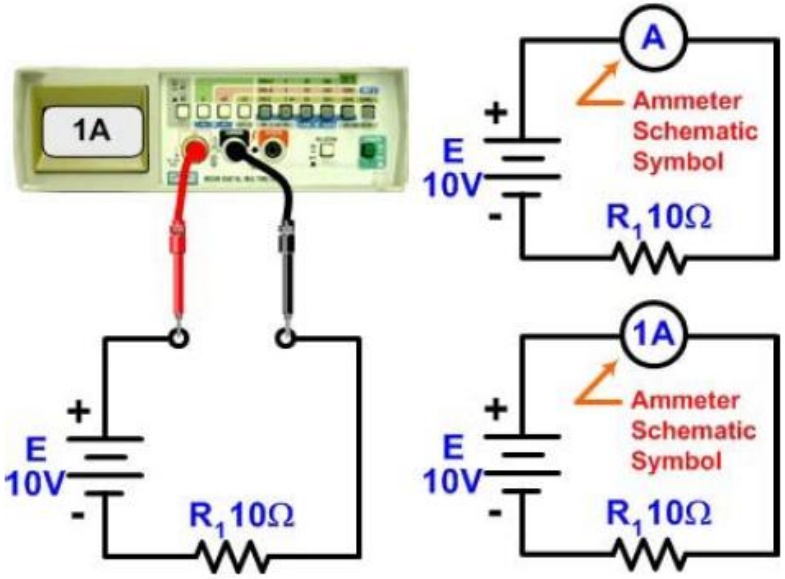
Ohmmeter II

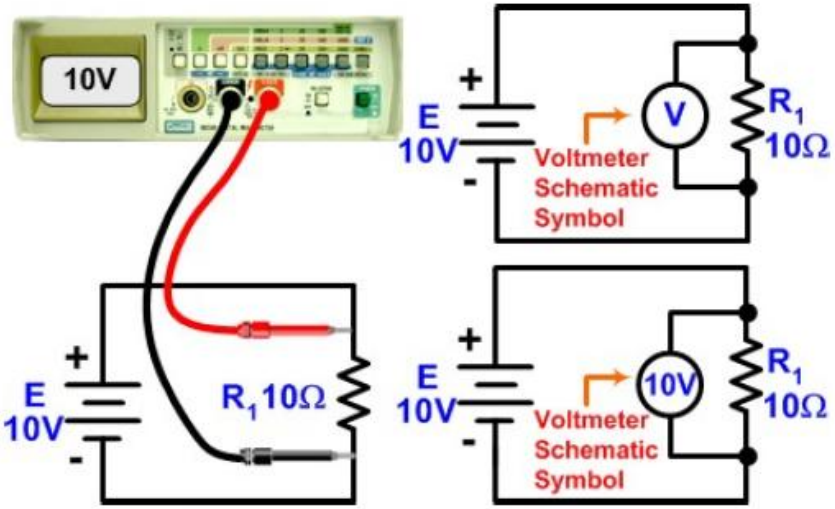
- Placing the meter in this circuit as shown will measure the combined resistance of R_1 and R_2.
  - Notice that R_1 and R_2 are connected at two points.
- To measure the resistance of R_1 only, R_1 must be isolated from R_2 by opening the path between R_1 and R_2.





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1A
A
Ammeter
Schematic
Symbol
E
10V
+
-
R1 10Ω
1A
Ammeter
Schematic
Symbol
E
10V
+
-
R1 10Ω


10V
E
10V
+
-
R1 10Ω
V
Voltmeter
Schematic
Symbol
R1
10Ω
E
10V
+
-
R1 10Ω
10V
Voltmeter
Schematic
Symbol
R1
10Ω

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